



2DE notes - Summary Differential equations, partial derivatives, vector geometry, matrix algebra, complex numbers, Taylor series.

Differential equations, partial derivatives, vector geometry, matrix algebra, complex numbers, Taylor series. (University of Cape Town)



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UNIVERSITY OF CAPE TOWN

MAM2000W - 2DE

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1

Introduction

Some Remarks Before we Start the Mathematics

You'll have noticed that these notes have big margins. Reading mathematics is definitely useful, but even more useful is reading and writing at the same time. The big margins are intended to hold your annotations. Whether you do this electronically, or take a more old-fashioned approach and print them out doesn't matter, but by the end of the course these should be your notes for 2DE, not mine.

There are different types of courses on differential equations. Some courses take a classification, methods and application approach. They introduce a type of differential equation, explain what it's good for and how to solve it, then move on to the next type. By the end of the course you'll know how to recognise and solve Bernoulli equations and Riccati equations and Cauchy-Euler equations and so on and so forth, but six months later it will all be gone¹. The alternative is to focus on depth, rather than breadth, and build a strong theoretical foundation. Since this is a pure mathematics course, that is what we are going to do. Certainly you won't learn how to solve as many different types of differential equation, but you will hopefully remember the foundations even if you don't remember all the details.

Although differential equations are used everywhere, from astrophysics to zoology², we're not going to focus too much on applications. We will occasionally make reference to physical problems, especially when they illustrate the theory in a particularly clear way, and will hopefully also have enough time to solve the Black-Scholes equation³, but we're not going to tie everything back to its real-world context.

There is a lot of computation in this course. Two important remarks. First, the theory that we develop can be applied to problems of arbitrary size, but the corresponding computations quickly become impractical to do by hand⁴ so we will limit ourselves to problems

¹ I speak from personal experience.

² And also in economics, finance and statistics.

³ Which got Fischer Black and Myron Scholes a Nobel prize in economics.

⁴ E.g. inverting a 4×4 matrix.

that can be tackled manually in a reasonable amount of time. This does mean that we are going to miss out on some interesting behaviour which only occurs in bigger problems. Secondly, you should always *think before you start computing*. A few minutes of thinking can often save an hour of frustrating computation.

Finally, there are a number of places in this course where we need to use complex numbers. There is nothing wrong with this, in fact it makes things much easier. We must, however, make sure to provide real-valued solutions to real-valued problems. Complex numbers introduced during the intermediate steps of a calculation should be neatly packed away once we're done with them.

The Start of the Mathematics

A **differential equation** (or DE) is an equation that relates a function to one or more of its derivatives, e.g.

$$\frac{dP}{dt} = kP(1 - P),$$

$$y'' + 4y' + y = 0,$$

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}.$$

Our first task is to classify differential equations. We're not going to attempt an exhaustive catalogue, but there are two main distinctions that we must make.

If the function in the differential equation is a function of a single variable, i.e. $P(t)$ or $y(x)$, then we call it an **ordinary differential equation** (or ODE). If the function depends on two or more variables, i.e. $u(x, t)$ or $q(x, y, z)$, and the differential equation includes partial derivatives with respect to two or more of these variables then we call it a **partial differential equation** (or PDE).

If a differential equation can be represented in the form

$$T(y) = f(x_1, x_2, \dots, x_n)$$

where T is a linear transformation between two function spaces then we call it a **linear differential equation**. If not we call it a **nonlinear differential equation**.

When we talk about linear transformations between two functions spaces, i.e. between two vector spaces of functions, we often call the mapping a **linear operator** rather than a linear transformation. We're not going to worry too much about what function spaces we're working in – we'll assume all the functions we are interested in live in $C^\infty[\mathbb{R}]$, i.e. they are smooth (infinitely differentiable) everywhere in $\mathbb{R}$. This is an entirely pragmatic decision. Functional analysis is

interesting, but it's difficult. If you decide to major in mathematics you'll get a taste of it in third year.

In this course we are going to spend all our time speaking about linear differential equations. It's not because nonlinear differential equations aren't interesting or useful, it's because the interesting and useful nonlinear differential equations are really hard to solve and each one needs to be solved differently. To paraphrase Tolstoy

"All linear equations are alike; each nonlinear equation is nonlinear in its own way."

The **order** of a differential equation is the highest order derivative that appears in the differential equation. The theory that we will develop for linear differential equations can be applied to n -th order problems but computation by hand quickly becomes infeasible.

A **solution** to a differential equation is a function that satisfies the differential equation. When the constant zero function is a solution to a differential equation we call it the **trivial solution**.

The existence and uniqueness of solutions to differential equations is guaranteed under certain conditions, and if this was a third-year or honours course in analysis we'd spend some time discussing these sorts of results:

THEOREM 1.1 (Picard-Lindelöf). *Let f be uniformly Lipschitz continuous. Then there is some $\epsilon > 0$ so that the initial value-problem*

$$y'(t) = f(t, y(t)), \quad y(t_0) = y_0$$

has a unique solution $y(t)$ in the interval $[t - \epsilon_0, t + \epsilon_0]$.

This is not a course in analysis. We can prove existence and uniqueness without too much trouble for linear ODEs (and we will later in the course) but nonlinear problems and PDEs require more serious machinery. If you choose to take MAM3000W next year I think you'll see Banach's Fixed Point theorem in 3MS at which point you can prove the theorem above, but for the rest of this course we will just assume that all the equations we're trying to solve have solutions.

Linear Independence and the Wronskian

In the following chapters we are often going to be interested in whether or not a particular set of functions is linearly independent. First a reminder of what this means.

DEFINITION. A set of functions $\{f_1, f_2, \dots, f_n\}$ is **linearly independent** on the domain D if the only solution to

$$\sum_{i=1}^n \alpha_i f_i = 0$$

is the trivial solution, i.e. all the coefficients $\alpha_i = 0$. Note that this must hold for all values of t in D . It's no good to find one set of coefficients that works when $t = 1$ and another when $t = 2$.

Checking linear independence using this definition is often difficult. Fortunately there is another method that works most of the time.

DEFINITION. The **Wronskian matrix** of n functions $\{f_1, f_2, \dots, f_n\}$, each differentiable $n - 1$ times on an interval I , is

$$M(t; f_1, f_2, \dots, f_n) = \begin{pmatrix} f_1(t) & f_2(t) & \cdots & f_n(t) \\ f_1'(t) & f_2'(t) & \cdots & f_n'(t) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(t) & f_2^{(n-1)}(t) & \cdots & f_n^{(n-1)}(t) \end{pmatrix}.$$

The **Wronskian** of these functions is the determinant of this matrix, i.e.

$$W(t; f_1, f_2, \dots, f_n) = \begin{vmatrix} f_1(t) & f_2(t) & \cdots & f_n(t) \\ f_1'(t) & f_2'(t) & \cdots & f_n'(t) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(t) & f_2^{(n-1)}(t) & \cdots & f_n^{(n-1)}(t) \end{vmatrix}.$$

THEOREM 1.2. If there is a point $t_0 \in I$ such that $W(t_0) \neq 0$ then the set $\{f_1, f_2, \dots, f_n\}$ is linearly independent on I .

Equivalently (the contrapositive) if the set $\{f_1, f_2, \dots, f_n\}$ is linearly dependent then $W(t) = 0$ for every $t \in I$.

Proof. We are given a set of functions $\{f_1, f_2, \dots, f_n\}$ each differentiable at least $n - 1$ times on an interval I . Consider the linear relation

$$\sum_{i=1}^n \alpha_i f_i = 0.$$

We must show that the only solution is the trivial one, i.e. all the coefficients α_i are zero. From this relation we can produce another by differentiating

$$\sum_{i=1}^n \alpha_i f_i' = 0.$$

And another by differentiating again

$$\sum_{i=1}^n \alpha_i f_i'' = 0.$$

In total we create n such relations by differentiating $n - 1$ times.

These relations can be compactly expressed as

$$\begin{pmatrix} f_1(t) & f_2(t) & \cdots & f_n(t) \\ f_1'(t) & f_2'(t) & \cdots & f_n'(t) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(t) & f_2^{(n-1)}(t) & \cdots & f_n^{(n-1)}(t) \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

If the Wronskian is nonzero at some point then the matrix on the left is invertible at that point. Multiply on both sides by its inverse to recover

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

The only solution is the trivial one. The set is linearly independent. $\square$

REMARK. It is a useful exercise to prove the theorem above but to work with the contrapositive statement, i.e. start with a linearly dependent set and show that the Wronskian must be zero everywhere.

REMARK. The converse is not true. If the Wronskian is zero everywhere the set of functions may still be linearly independent, as the following example demonstrates.

EXAMPLE. Is the set $\{t^2, t|t|\}$ linearly independent or linearly dependent on $\mathbb{R}$?

Let's start with the Wronskian. In this case we have

$$W(t) = \begin{vmatrix} t^2 & t|t| \\ 2t & 2|t| \end{vmatrix} = 2t^2|t| - 2t^2|t| = 0.$$

A nonzero Wronskian implies linear independence. A zero Wronskian implies nothing. We have to go back to the definition.

Consider the linear relation

$$\alpha t^2 + \beta t|t| = 0.$$

This should hold for every real t . In particular it should hold for $t = 1$ and $t = -1$.

$$\begin{aligned} t = 1 &\Rightarrow \alpha + \beta = 0, \\ t = -1 &\Rightarrow \alpha - \beta = 0. \end{aligned}$$

From this we conclude that $\alpha = \beta = 0$ and so the set is linearly independent.

Make sure you understand what this example tells you about the theorem above.

EXAMPLE. Show that the set $\{1, t, t^2, \dots, t^n\}$ is linearly independent.

$$\begin{aligned}
 W(t) &= \begin{vmatrix} 1 & t & t^2 & \cdots & t^n \\ 0 & 1 & 2t & \cdots & nt^{n-1} \\ 0 & 0 & 2 & \cdots & n(n-1)t^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & n! \end{vmatrix} \\
 &= \prod_{k=0}^n k! \\
 &\neq 0.
 \end{aligned}$$

The set is linearly independent.

In the following chapters we're not going to be looking at linear independence of arbitrary sets of functions, but rather at linear independence of sets of functions where all of those functions are solutions to the same linear differential equation. In this case we have the following, stronger result.

THEOREM 1.3. *Let $f_1, f_2, \dots, f_n$ all be solutions to the same n -th order linear homogeneous ordinary differential equation.*

If and only if there is a point $t_0 \in I$ such that $W(t_0) \neq 0$ then the set $\{f_1, f_2, \dots, f_n\}$ is linearly independent on I .

Proof. The proof (for $n = 2$) is a tutorial problem. □

REMARK. Compare theorems 1.2 and 1.3. Make sure you understand the difference between them. What is the contrapositive of the statement in 1.3?

2

Linear Ordinary Differential Equations

An **n-th order linear ordinary differential** has the form

$$a_n(t) \frac{d^n y}{dt^n} + a_{n-1}(t) \frac{d^{n-1} y}{dt^{n-1}} + \dots + a_0(t)y = f(t)$$

where the function $a_n \neq 0$. If $f = 0$ we call this differential equation **homogeneous** otherwise we call it **nonhomogeneous**.

When the functions a_i are constants this is called a **constant coefficient** differential equation. We will spend a lot of time speaking about this special case. In general its hard to solve linear differential equations if a_i are arbitrary functions.

The following notation (and the corresponding conceptual shift) is useful when dealing with linear ordinary differential equations. Denote by D the linear operator $D : C^\infty \rightarrow C^\infty$ which sends $f \mapsto \frac{df}{dt}$ and I the identity operator which sends $f \mapsto f$.

The linear differential equation written above can be rewritten as

$$T(y) = f \tag{2.1}$$

where the linear operator¹ T is

$$T = a_n D^n + a_{n-1} D^{n-1} + \dots + a_0 I.$$

¹ You should convince yourselves that T is really linear as claimed.

This way of writing down a differential equation might remind you of something you've seen before. If A is an $m \times n$ matrix, $\mathbf{x}$ an $n \times 1$ vector and $\mathbf{b}$ an $m \times 1$ vector then

$$A\mathbf{x} = \mathbf{b}$$

has the same structure as (2.1), i.e. a linear transformation acting on one vector generates another, and many of the results about solutions to systems of linear equations have analogs when it comes to solving linear differential equations.

Homogeneous Differential Equations with Constant Coefficients

Linear differential equations with constant coefficients can all be solved. We start with the first order case where the behaviour is very simple, then the second order case where we get all the different types of behaviour for linear differential equations, and finally we extend this to the n -th order case which we will see can be solved one piece at a time.

First Order Differential Equations

A first order, linear, homogeneous, constant coefficient differential equation has the form

$$a_1 \frac{dy}{dt} + a_0 y = 0.$$

This is not the most useful way to write it. If we let $\lambda = -\frac{a_0}{a_1}$ then we can rewrite this as

$$(D - \lambda I)y = 0 \quad (2.2)$$

or

$$Dy = \lambda y \quad (2.3)$$

depending on whether we want to emphasise its connection to higher order problems (2.2) or that it's an eigenvalue problem (2.3). However we write it the following is true

THEOREM 2.1. *The solutions to the differential equation $Dy = \lambda y$ are exponentials of the form $\alpha e^{\lambda t}$. The initial value problem $Dy = \lambda y$ with $y(0) = y_0$ has a unique solution $y = y_0 e^{\lambda t}$.*

Proof. Clearly the exponentials $\alpha e^{\lambda t}$ are solutions since

$$\begin{aligned} D(\alpha e^{\lambda t}) &= \alpha D(e^{\lambda t}) \\ &= \alpha \lambda e^{\lambda t} \\ &= \lambda (\alpha e^{\lambda t}). \end{aligned}$$

We should show that these are the only solutions. Let f be an arbitrary solution to $Dy = \lambda y$. Note that

$$\begin{aligned} D(e^{-\lambda t} f) &= -\lambda e^{-\lambda t} f + e^{-\lambda t} Df \\ &= -\lambda e^{-\lambda t} f + e^{-\lambda t} \lambda f \\ &= 0 \end{aligned}$$

which means that $e^{-\lambda t} f = \alpha \in \mathbb{R}$ or $f = \alpha e^{\lambda t}$. The solutions really must all be exponentials (or in the special case where $\lambda = 0$, constant functions) and if there is an initial condition this uniquely selects one of those exponentials as the solution. $\square$

This analysis has been done for *real* λ . As long as we are dealing with linear differential equations with real coefficients, operators $(D - \lambda I)$ where λ has a non-zero imaginary component will only appear as part of a pair² $(D - \lambda I)(D - \bar{\lambda} I)$. We discuss how to deal with this situation in the following sections.

² This is just like a polynomial with real coefficients. If it has a root with non-zero imaginary components that root appears as part of a conjugate pairs.

Second Order Differential Equations

A second order, linear, homogeneous, constant coefficient differential equation has the form

$$a_2 \frac{d^2 y}{dt^2} + a_1 \frac{dy}{dt} + a_0 y = 0.$$

Since a_2 is nonzero we can safely divide through and so we can restrict ourselves to equations

$$D^2 y + a_1 D y + a_0 y = 0$$

without loss of generality.

Associated to every constant coefficient differential equation is an **auxiliary polynomial** $p(\lambda)$. This polynomial can be determined in two different ways. The first is to assume (based on what we know about first order equations) that the solution y has the form $y = e^{\lambda t}$. Substitution into the equation gives us

$$\begin{aligned} D^2 (e^{\lambda t}) + a_1 D (e^{\lambda t}) + a_0 e^{\lambda t} &= 0, \\ \lambda^2 e^{\lambda t} + a_1 \lambda e^{\lambda t} + a_0 e^{\lambda t} &= 0, \\ \underbrace{\lambda^2 + a_1 \lambda + a_0}_{p(\lambda)} &= 0. \end{aligned}$$

Alternatively, we might recognise that the differential equation itself can be rewritten as a polynomial in D acting on y , i.e.

$$\underbrace{(D^2 + a_1 D + a_0 I)}_{p(D)} y = 0.$$

This polynomial in D has the same coefficients as the auxiliary polynomial, all we need to do is replace D with λ .

The polynomial $p(D)$ can be factorised in exactly the same way that $p(\lambda)$ can be factorised – this observation will be very useful when we discuss the n -dimensional case. For now we are interested in the roots of the quadratic $p(\lambda)$. There are three different possibilities to consider

1. $p(\lambda)$ has two distinct real roots,
2. $p(\lambda)$ has a repeated real root,

3. $p(\lambda)$ has a complex conjugate pair of eigenvalues.

If the auxiliary polynomial $p(\lambda)$ has two distinct real roots λ_1 and λ_2 then it can be factorised as $p(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2)$. The differential equation can be likewise factored and written as

$$(D - \lambda_1 I)(D - \lambda_2 I)y = 0.$$

It is easy enough to check that the functions $y_1 = e^{\lambda_1 t}$ and $y_2 = e^{\lambda_2 t}$ are solutions to the differential equation. Although we haven't shown it yet we will see that a second order differential equation has a two-dimensional solution space (Theorem 2.6). These two linearly independent solutions form a basis for the solution space. We call such a basis, i.e. $\{y_1, y_2\}$, a **fundamental set** and the **general solution** to this differential equation has the form

$$y = \alpha y_1 + \beta y_2 = \alpha e^{\lambda_1 t} + \beta e^{\lambda_2 t}.$$

If $p(\lambda)$ has one repeated real root λ_1 it can be factorised as $p(\lambda) = (\lambda - \lambda_1)^2$ and the differential equation can be rewritten like

$$(D - \lambda_1 I)^2 y = 0.$$

This is a second order equation, so we need two linearly independent solutions for the fundamental set. The first is $y_1 = e^{\lambda_1 t}$. The second is $y_2 = te^{\lambda_1 t}$. It's not obvious why this should work, but you can check that it does by substituting in to the differential equation. It's also not too hard to show that these two solutions are linearly independent. We will discuss how we know the second solution has this form a little later (see Theorem 2.2). The general solution in this case is

$$y = \alpha y_1 + \beta y_2 = \alpha e^{\lambda_1 t} + \beta te^{\lambda_1 t}.$$

Finally, $p(\lambda)$ may have a complex conjugate pair of roots $\lambda_{1,2} = \mu \pm \omega i$. It can be factorised either as $p(\lambda) = (\lambda - \lambda_1)(\lambda - \bar{\lambda}_1)$ or written as $p(\lambda) = (\lambda - \mu)^2 + \omega^2$, depending on what aspect of the problem we want to emphasise. Similarly, the differential equation could be written either as

$$(D - \lambda_1 I)(D - \bar{\lambda}_1 I)y = 0$$

or

$$((D - \mu I)^2 + \omega^2 I)y = 0.$$

Two linearly independent solutions to this differential equation are $y_1 = e^{\mu t} \cos(\omega t)$ and $y_2 = e^{\mu t} \sin(\omega t)$. The general solution is

$$y = \alpha e^{\mu t} \cos(\omega t) + \beta e^{\mu t} \sin(\omega t).$$

It is not obvious where these solutions come from. Although it's good that the solutions are completely real (which makes sense

because the differential equation had real coefficients) it's not clear how we deal with the complex roots. It will happen again later in the course that we need to combine complex quantities in some combination to produce a real output so make sure you understand the following.

The solutions to our differential equation are exponentials of the form $e^{\lambda t}$. In this case $e^{\mu t} e^{i\omega t}$ and $e^{\mu t} e^{-i\omega t}$. The general solution is therefore

$$y = Ce^{\mu t} e^{i\omega t} + De^{\mu t} e^{-i\omega t}.$$

Note that if z is complex, $z + \bar{z}$ is real. We therefore don't take an arbitrary linear combination, but restrict ourselves to $D = \bar{C}$. In this case

$$\begin{aligned} y &= Ce^{\mu t} e^{i\omega t} + \bar{C}e^{\mu t} e^{-i\omega t} \\ &= e^{\mu t} (Ce^{i\omega t} + \bar{C}e^{-i\omega t}) \\ &= e^{\mu t} (C \cos(\omega t) + iC \sin(\omega t) + \bar{C} \cos(\omega t) - i\bar{C} \sin(\omega t)) \\ &= e^{\mu t} \left(\underbrace{C \cos(\omega t) + \bar{C} \cos(\omega t)}_{2\Re(C) \cos(\omega t)} + i \underbrace{(C \sin(\omega t) - \bar{C} \sin(\omega t))}_{2i\Im(C) \sin(\omega t)} \right) \\ &= 2\Re(C)e^{\mu t} \cos(\omega t) - 2\Im(C)e^{\mu t} \sin(\omega t) \\ &= \alpha e^{\mu t} \cos(\omega t) + \beta e^{\mu t} \sin(\omega t) \end{aligned}$$

These different types of behaviour can be very neatly illustrated by a physical model. Damped harmonic oscillators³ have the form

$$y'' + \gamma y' + \omega^2 y = 0$$

where $\gamma \geq 0$ describes the strength of a damping force (like friction, air resistance, vibration etc.) and ω is a real number related to the strength of the restoring force (like gravity pulling a pendulum back to its rest position, or the springs in a shock absorber pushing the shock arm back out after passing over a bump in the road). How does this system behave? It depends on the strength of the damping force relative to the restoring force.

The auxiliary polynomial is $p(\lambda) = \lambda^2 - \gamma\lambda + \omega^2$. The roots of this polynomial are at

$$\lambda_{1,2} = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega^2}}{2}.$$

Notice that there is a **critical value** $\gamma_c = 2\omega$ where the discriminant changes sign.

1. If $\gamma = 0$ the system is **undamped**. The roots $\lambda_{1,2}$ are pure imaginary and the systems oscillates periodically.

³ If you're not interested in the physics, that's fine. Just look at how the solutions to this differential equations change as the relative values of γ and ω change.

2. If $0 < \gamma < \gamma_c$ the system is **underdamped**. The roots are a complex conjugate pair and the system exhibits attenuated oscillation (i.e. the amplitude of the oscillations decreases with time).

Underdamping is not necessarily good or bad. In an old-fashioned grandfather clock powered by a pendulum we'd like damping to be minimised so that it keeps time for as long as possible. In a car's suspension underdamping is terrible since passing over a bump would lead to a bouncy ride.

3. If $\gamma = \gamma_c$ the system is **critically damped**. There is a single repeated real root. When a critically damped system is disturbed it returns to the rest position faster than an underdamped or overdamped system. A good suspension is critically damped.
4. If $\gamma_c < \gamma$ the system is **overdamped**. There are two real roots, one larger than the root in the critically damped case and one smaller. This means that the system takes a long time to return to the rest position after it's been disturbed. A door with a stiff automatic lever to close it is an example of an overdamped system.

Below we plot representative behaviour for an undamped, underdamped, critically damped and overdamped systems. Specifically, we plot the solutions to

$$y'' + \gamma y' + y = 0, y(0) = 1, y'(0) = 0$$

for $\gamma = 0, 1, 2, 3$. Those solutions are

$$\gamma = 0: \quad y = \cos(t)$$

$$\gamma = 1: \quad y = e^{-\frac{1}{2}t} \cos\left(\frac{\sqrt{3}}{2}t\right) + \frac{1}{\sqrt{3}}e^{-\frac{1}{2}t} \sin\left(\frac{\sqrt{3}}{2}t\right)$$

$$\gamma = 2: \quad y = e^{-t} + te^{-t}$$

$$\gamma = 3: \quad y = \frac{5 + 3\sqrt{5}}{10}e^{-\frac{3-\sqrt{5}}{2}t} + \frac{5 - 3\sqrt{5}}{10}e^{-\frac{3+\sqrt{5}}{2}t}$$

and their behaviour is plotted below in Figure 2.1.

At this stage you should have two outstanding questions about the material we have just covered. How do we know that we really have found all the solutions to these differential equations, i.e. is it really the case that a second-order linear differential equation has a two-dimensional solution space as claimed? Why do we use $te^{\lambda t}$ as a second linearly independent solution when we have a repeated root? Both of these issues are addressed in the next section.

n-th Order Differential Equations

Solving n -th order linear differential equations with constant coefficients is surprisingly easy: the solution to the differential equation

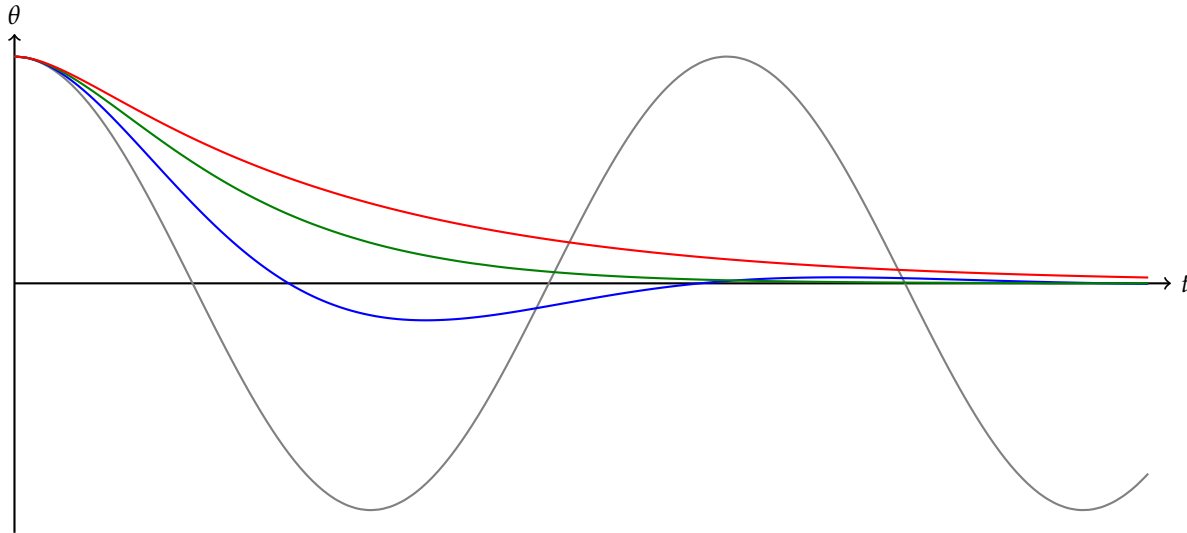


Figure 2.1: Behaviour of an undamped (grey), underdamped (blue), critically damped (green) and overdamped (red) system.

can be immediately produced if the roots of its auxiliary polynomial are known. We're going to jump in with a few examples, using what we know about the second-order equations to guide us, and once we've established how to solve the equations we'll discuss in detail why this method works.

EXAMPLE. Write down the fundamental set and hence the general solution for the differential equation

$$y''' - y'' + 2y' - 2y = 0.$$

The first step is to calculate the auxiliary polynomial. As before we can do this by assuming that the solution has the form $y = e^{\lambda t}$. For this equation the auxiliary polynomial is

$$p(\lambda) = \lambda^3 - \lambda^2 + 2\lambda - 2.$$

Next we look for the roots of this polynomial. It's not as easy as finding the roots of a quadratic, but we may notice by inspection that $\lambda_1 = 1$ works. We can thus factorise $p(\lambda)$ as

$$p(\lambda) = (\lambda - 1)(\lambda^2 + 2).$$

The remaining two roots are therefore $\lambda_{2,3} = \pm\sqrt{2}i$. The contribution of the roots to the solution is analogous to the second-order case. Of course it's not completely the same: we will now end up with *three* linearly independent solutions corresponding to the *three* roots of the *third-order* polynomial associated with a *third-order* differential equation⁴. The fundamental set is

$$\left\{ e^t, \cos(\sqrt{2}t), \sin(\sqrt{2}t) \right\}$$

⁴ I'm belabouring this point, the "three-ness" of everything, because it's important.

and the general solution is

$$y = \alpha e^t + \beta \cos(\sqrt{2}t) + \gamma \sin(\sqrt{2}t).$$

EXAMPLE. Write down the fundamental set and hence the general solution for the differential equation

$$(D + I)^3(D^2 + 2I)y = 0.$$

Again, we start by calculating the auxiliary polynomial. If we remember that linear constant coefficient differential equations can be expressed as polynomials in D , i.e. $p(D)y = 0$ it's quick to see that the corresponding auxiliary polynomial is

$$p(\lambda) = (\lambda + 1)^3(\lambda^2 + 2)$$

which has roots $\lambda_{1,2,3} = -1$ and $\lambda_{4,5} = \sqrt{2}i$. We haven't discussed what to do with roots that are repeated more than twice, but we could make the following guess for the fundamental set⁵

$$\left\{ e^{-t}, te^{-t}, t^2e^{-t}, \cos(\sqrt{2}t), \sin(\sqrt{2}t) \right\}.$$

The general solution would therefore be

$$y = \alpha_1 e^{-t} + \alpha_2 t e^{-t} + \alpha_3 t^2 e^{-t} + \alpha_4 \cos(\sqrt{2}t) + \alpha_5 \sin(\sqrt{2}t).$$

It really is very easy to solve this type of differential equations. Find the auxiliary polynomial, find the roots, write down the solution. Explaining why this works is a little harder. We need to show the following.

1. Any n -th order linear homogeneous differential equation can be rewritten as a composition of operators of the form $(D - \lambda_i I)^{k_i}$ sending y to 0. Note that $\sum k_i = n$.
2. The differential equation $(D - \lambda I)^k y = 0$ has k linearly independent solutions.
3. Any linear combination of solutions to the individual differential equations is a solution to the original n -th order differential equation.
4. Every solution to an n -th order differential equation can be written as a linear combination of solutions to its constituent parts, i.e. a n -th order differential equation has an n -dimensional solution space.

The first point is a consequence of the fundamental theorem of algebra. Although you've only seen it in the context of factoring polynomials in variable like x or λ it holds here as well. Note

⁵ Based on what we know about the second-order case.

that if the coefficients in the differential equation are real then a factor $(D - \lambda I)$ with complex λ will always appear as part of the pair $(D - \lambda I)(D - \bar{\lambda} I)$.

The second point requires a little bit of work.

THEOREM 2.2. *The differential equation*

$$(D - \lambda I)^k y = 0$$

has k linearly independent solutions

$$\{e^{\lambda t}, te^{\lambda t}, t^2 e^{\lambda t}, \dots, t^{k-1} e^{\lambda t}\}.$$

Proof. We start by showing that these k functions really are all solutions. We proceed by using mathematical induction.

For the base case, $k = 1$, refer back to Theorem 2.1.

Assume that $e^{\lambda t}, te^{\lambda t}, \dots, t^{j-1} e^{\lambda t}$ solve $(D - \lambda I)^j y = 0$.

For $k = j + 1$ we have

$$\begin{aligned} (D - \lambda I)^{j+1}(t^j e^{\lambda t}) &= (D - \lambda I)^j (D - \lambda I)(t^j e^{\lambda t}) \\ &= (D - \lambda I)^j (jt^{j-1} e^{\lambda t} + \lambda t^j e^{\lambda t} - \lambda t^j e^{\lambda t}) \\ &= j(D - \lambda I)^j (t^{j-1} e^{\lambda t}) \\ &= 0. \end{aligned}$$

Therefore $t^j e^{\lambda t}$ is a solution to $(D - \lambda I)^{j+1} y = 0$. Fairly obviously all the solutions to $(D - \lambda I)^j y = 0$ are also solutions to $(D - \lambda I)^{j+1} y = 0$. What remains is to show that these are linearly independent solutions.

We consider the equation

$$\alpha_1 e^{\lambda t} + \alpha_2 t e^{\lambda t} + \alpha_3 t^2 e^{\lambda t} + \dots + \alpha_k t^{k-1} e^{\lambda t} = 0.$$

After taking out a common factor we have

$$e^{\lambda t} (\alpha_1 + \alpha_2 t + \alpha_3 t^2 + \dots + \alpha_k t^{k-1}) = 0.$$

The factor $e^{\lambda t}$ is always nonzero, so we are looking for α_i that satisfy

$$\alpha_1 + \alpha_2 t + \alpha_3 t^2 + \dots + \alpha_k t^{k-1} = 0.$$

We know that the only solution to this is the trivial one, $\alpha_i = 0$, because the functions $1, t, t^2, \dots, t^{k-1}$ are linearly independent⁶. Therefore the set

$$\{e^{\lambda t}, te^{\lambda t}, t^2 e^{\lambda t}, \dots, t^{k-1} e^{\lambda t}\}$$

is linearly independent. □

⁶ We showed this earlier when discussing the Wronskian

REMARK. As we will shortly see (Theorem 2.6) there are exactly k linearly independent solutions and the kernel of $(D - \lambda I)^k$ is k -dimensional.

REMARK. If $\lambda = \mu + \omega i$ is a complex number then the k independent solutions have nonzero imaginary components. However, there will be a corresponding k independent solutions to $(D - \bar{\lambda}I)^k$, and by taking appropriate linear combinations, like we did with the second-order differential equations, we can eliminate the imaginary components and recover $2k$ linearly independent real solutions

$$\left\{ e^{\mu t} \cos(\omega t), e^{\mu t} \sin(\omega t), t e^{\mu t} \cos(\omega t), t e^{\mu t} \sin(\omega t), \dots, t^{k-1} e^{\mu t} \cos(\omega t), t^{k-1} e^{\mu t} \sin(\omega t) \right\}.$$

THEOREM 2.3. *If y_p and y_q are non-trivial solutions to $(D - \lambda_p I)^{k_p} y = 0$ and $(D - \lambda_q I)^{k_q} y = 0$ (where $\lambda_p \neq \lambda_q$) then the set $\{y_p, y_q\}$ is linearly independent.*

Proof. We prove the case where $k_p = k_q = 1$. You should try to extend this proof to the more general case. There are some guidelines at the bottom of this proof.

Consider two non-trivial solutions y_p and y_q . We look for solutions to

$$\alpha y_p + \beta y_q = 0.$$

Apply the operator $(D - \lambda_p I)$ to this equation. The first term disappears since y_p is in its kernel, so

$$\begin{aligned} \beta(D - \lambda_p I)y_q &= 0 \\ \beta(\lambda_q - \lambda_p)y_q &= 0. \end{aligned}$$

The function y_q is nontrivial and the term $(\lambda_q - \lambda_p) \neq 0$ so $\beta = 0$. A similar argument shows that $\alpha = 0$ and so we have linear independence.

To extend this argument consider $k_p \in \mathbb{N}$ and $k_q = 1$ as the base case, and then make an inductive argument on k_q , i.e. assume it holds for $k_q = n$ and consider $k_q = n + 1$.

□

The final point requires more than just a little work. We start with some lemmas.

LEMMA 2.4. *The operator $(D - \lambda I) : C^\infty \rightarrow C^\infty$ is onto (surjective / an epimorphism).*

Proof. Let $v \in C^\infty$. Since v is smooth it is continuous and hence integrable. Define another smooth function u by

$$u = \left(\int^t v(s) e^{-\lambda s} ds \right) e^{\lambda t}.$$

Then

$$(D - \lambda I)u = (ve^{-\lambda t})e^{\lambda t} + \lambda \left(\int^t ve^{-\lambda s} ds \right) e^{\lambda t} - \lambda \left(\int^t ve^{-\lambda s} ds \right) e^{\lambda t} = v$$

so $(D - \lambda I)$ is onto as claimed. $\square$

LEMMA 2.5. *Let V be a vector space. If T and U are both linear maps $V \rightarrow V$ with finite-dimensional kernels and U is onto then*

$$\dim(\ker(TU)) = \dim(\ker(T)) + \dim(\ker(U))$$

Proof. Let $B_U = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ be a basis for $\ker(U)$ (which is therefore p -dimensional) and $B_T = \{\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_q\}$ be a basis for $\ker(T)$ (therefore q -dimensional).

Since U is onto we can find vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_q$ that are mapped to the basis vectors $\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_q$ by U , i.e. $U(\mathbf{v}_1) = \mathbf{t}_1$ and so on.

Basis vectors must be nonzero, and the vectors $\mathbf{u}_i$ lie in the kernel of U , so we have $\mathbf{u}_i \neq \mathbf{v}_j$ for any i, j .

The set

$$B = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_q\}$$

therefore has $p + q$ elements and if we can show that it is a basis for the kernel of TU then we will have shown that the kernel of TU is $p + q$ -dimensional as required.

First we show that B is a linearly independent set. Consider the linear relation

$$\sum_{i=1}^p \alpha_i \mathbf{u}_i + \sum_{j=1}^q \beta_j \mathbf{v}_j = \mathbf{0}. \quad (2.4)$$

Apply the map U to this linear relation. Since the vectors $\mathbf{u}_i$ lie in the kernel of U and the vectors $\mathbf{v}_j$ map to $\mathbf{t}_j$ we recover

$$\sum_{j=1}^q \beta_j \mathbf{t}_j = \mathbf{0}.$$

The coefficients β_j are all zero since the set B_T must be linearly independent (it's a basis). This reduces equation 2.4 to

$$\sum_{i=1}^p \alpha_i \mathbf{u}_i = \mathbf{0}$$

which implies that all the coefficients α_i are zero (by similar reasoning). The set B is linearly independent.

Finally we show that $\langle B \rangle = \ker(TU)$, i.e. $\langle B \rangle \subseteq \ker(TU)$ and $\ker(TU) \subseteq \langle B \rangle$. The first inclusion is easy since

$$TU \left(\sum_{i=1}^p \alpha_i \mathbf{u}_i + \sum_{j=1}^q \beta_j \mathbf{v}_j \right) = T \left(\sum_{j=1}^q \beta_j \mathbf{t}_j \right) = \mathbf{0}.$$

The second is harder. Let $\mathbf{w} \in \ker(TU)$. If $\mathbf{w} \in \langle B_U \rangle$ we are immediately done, so assume otherwise, i.e.

$$U(\mathbf{w}) = \mathbf{z} \neq \mathbf{o}.$$

But $TU(\mathbf{w}) = \mathbf{o}$ so $T(\mathbf{z}) = \mathbf{o}$ which means $\mathbf{z} \in \langle B_T \rangle$.

$$\begin{aligned}\mathbf{z} &= \sum_{j=1}^q \beta_j \mathbf{t}_j \\ U(\mathbf{w}) &= \sum_{j=1}^q \beta_j U(\mathbf{v}_j) \\ U\left(\mathbf{w} - \sum_{j=1}^q \beta_j \mathbf{v}_j\right) &= \mathbf{o} \\ \mathbf{w} - \sum_{j=1}^q \beta_j \mathbf{v}_j &= \sum_{i=1}^p \alpha_i \mathbf{u}_i \\ \mathbf{w} &= \sum_{i=1}^p \alpha_i \mathbf{u}_i + \sum_{j=1}^q \beta_j \mathbf{v}_j\end{aligned}$$

The proof is complete since this inclusion gives equality which (together with the linear independence) makes B a basis of $p + q$ elements for $\ker(TU)$. $\square$

THEOREM 2.6. *An n -th order, linear, homogeneous, constant coefficient differential equation has an n -dimensional solution space.*

Proof. The proof is by induction. The heavy lifting is going to be done by Lemma 2.5.

The base case, $n = 1$, was discussed earlier (Theorem 2.1).

Assume that it is true for $n = k$, i.e. a k -th order linear, homogeneous, constant coefficient differential equation has a k -dimensional solution space.

Consider $n = k + 1$. The constant coefficient differential equation can be factorised and written as

$$(D - \lambda_1 I)(D - \lambda_2 I) \cdots (D - \lambda_k I)(D - \lambda_{k+1} I)y = 0.$$

Alternatively, we can think of it as

$$T(D - \lambda_{k+1} I)y = 0$$

where the operator T is the composition of the first k operators in the previous line. Since $(D - \lambda_{k+1} I)$ is onto (Lemma 2.4) the conditions for Lemma 2.5 are satisfied. Then the $k + 1$ -th order linear, homogeneous, constant coefficient differential equation has a $k + 1$ -dimensional solution space.

By the principle of mathematical induction the result holds. $\square$

REMARK. This result holds for all linear homogeneous differential equations, not just ones with constant coefficients. We've proved it for only the constant coefficient case because that's the best we can do with the tools we have at the moment.

Not all differential equations are homogeneous. We discuss the nonhomogeneous case below. The solutions to homogeneous differential equations play a big role in solving nonhomogeneous equations.

Nonhomogeneous Differential Equations with Constant Coefficients

We're going to discuss two different approaches (the annihilator method and variation of parameters) to solving nonhomogeneous differential equations. Each of them has their own strengths and weaknesses, and you should also think about which is more appropriate for the particular problem you have to solve. Before we get to these approaches we should make an important general remark about the solution to nonhomogeneous differential equations.

Every linear nonhomogeneous differential equation (not necessarily with constant coefficients)

$$T(y) = f(t)$$

(where f is not identically zero) has an associated homogeneous differential equation

$$T(y) = 0.$$

THEOREM 2.7. *The **general solution** to a nonhomogeneous differential equation is the sum of y_0 , the solution to the associated homogeneous equation, in this context called the **complementary solution**, and any solution to the nonhomogeneous differential equation, y_p , called the **particular solution**.*

$$y = y_0 + y_p$$

REMARK. This result is analogous to one you've seen before in 2LA about solutions to systems of algebraic equations (I believe it's Theorem 43, part 3 in those notes). We're not going to prove it here but you should make sure that you can extend the argument from linear algebraic equations to linear differential equations.

REMARK. The differential equation $T(y) = f_1 + f_2 + \dots + f_n$ has the general solution

$$y = y_0 + y_{p_1} + y_{p_2} + \dots + y_{p_n}$$

where each y_{p_j} is a particular solution to $T(y) = f_j$.

The first method we discuss for solving nonhomogeneous equations, the annihilator method, solves for the complementary and particular solution at the same time. The second, variation of parameters, is a method for determining a particular solution from the complementary solution.

The Annihilator Method

We consider a nonhomogeneous differential equation

$$T(y) = f.$$

The idea behind the annihilator method is the following. If the function f lies in the kernel of another operator A , called the **annihilator**, we can apply A to recover the homogeneous differential equation

$$(AT)(y) = A(f) = 0.$$

This homogeneous differential equation can be solved. The resulting solution to this higher-order differential equation is substituted into the original differential equation and some of the undetermined coefficients in the general solution are fixed. The resulting expression for y is easily recognised as being the sum of a complementary and particular solution. This sounds a little abstract so here are some examples to make things concrete.

EXAMPLE. Solve the differential equation

$$y'' + y = e^t. \quad (2.5)$$

First we notice that the function e^t , the nonhomogeneous part of this differential equation, is a solution to the differential equation $(D - I)y = 0$. We therefore let $A = (D - I)$.

This is not the only annihilator that would work, but it is the simplest. There is no need to take $A = (D - I)^2$ or $A = (D - I)(D + 3I)$. We should choose the minimal annihilator. Having made that choice we apply the annihilator to equation 2.5

$$\begin{aligned} (D^2 + I)y &= e^t \\ (D - I)(D^2 + I)y &= (D - I)e^t \\ (D - I)(D^2 + I)y &= 0 \end{aligned}$$

and recover a homogeneous, constant coefficient equation. The auxiliary polynomial is

$$p(\lambda) = (\lambda - 1)(\lambda^2 + 1)$$

and so the general solution to this equation is

$$y = Ae^t + B \cos(t) + C \sin(t).$$

Note that the differential equation we started with was second-order, but the solution above has three unknown constants in it. We resolve this by substituting the solution back into equation 2.5.

$$(D^2 + I)(Ae^t + B \cos(t) + C \sin(t)) = e^t.$$

The $B \cos(t)$ and $C \sin(t)$ terms are solutions to $(D^2 + I)y = 0$ so they disappear when operated on. We have

$$\begin{aligned}(D^2 + I)Ae^t &= e^t \\ A(D^2 e^t) + Ae^t &= e^t \\ Ae^t + Ae^t &= e^t\end{aligned}$$

so the constant $A = \frac{1}{2}$. The general solution to equation 2.5 is therefore

$$y = \underbrace{B \cos(t) + C \sin(t)}_{y_0} + \underbrace{\frac{1}{2}e^t}_{y_p}.$$

Solving nonhomogeneous equations using the annihilator method requires that you are fluent in solving homogeneous equations so that you can easily recognise which differential operator could serve as an annihilator. Some more examples.

EXAMPLE. Solve the differential equation

$$y' + 2y = te^{-t}. \quad (2.6)$$

Here the appropriate annihilator is $A = (D + I)^2$. Apply to equation 2.6 and recover a homogeneous equation

$$(D + I)^2(D + 2I)y = 0.$$

The solution to this homogeneous equation is

$$y = Ae^{-t} + Bte^{-t} + Ce^{-2t}.$$

Substitute into equation 2.6 to eliminate some of the constants. After some calculation you should find that

$$(A + B)e^{-t} + Bte^{-t} = te^{-t}.$$

By comparing coefficients we see that $A = -1$ and $B = 1$. The general solution to this nonhomogeneous differential equation is

$$y = \underbrace{Ce^{-2t}}_{y_0} + \underbrace{te^{-t} - e^{-t}}_{y_p}.$$

EXAMPLE. Solve the differential equation

$$y'' + y' - 2y = e^t. \quad (2.7)$$

Here the appropriate annihilator is $A = (D - I)$. Apply to equation 2.7 and recover a homogeneous equation

$$(D - I)^2(D + 2I)y = 0.$$

The solution to this homogeneous equation is

$$y = Ae^t + Bte^t + Ce^{-2t}.$$

Substitute into equation 2.7 to eliminate some of the constants. After some calculation you should find that

$$3Be^t = e^t.$$

so $B = \frac{1}{3}$. The general solution to this nonhomogeneous differential equation is

$$y = \underbrace{Ae^t + Ce^{-2t}}_{y_0} + \underbrace{\frac{1}{3}te^t}_{y_p}.$$

The annihilator method is straightforward. It requires only that we understand the homogeneous case well to solve the inhomogeneous case. Unfortunately it suffers from two major problems. It's not possible to annihilate every function with a polynomial in D , e.g. $\tan(t)$ or $\ln(t)$ and it can't be generalised to differential equations without constant coefficients. Variation of parameters solves both of these problems.

Variation of Parameters

Variation of parameters works on n -th order differential equations but doing the calculations by hand becomes completely impractical. We will discuss the second order case to start and do a few second order examples. Once we understand the simple case we move to the n -th order case and do one example of a third order equation.

Consider the nonhomogeneous linear differential equation

$$a_2(t)y'' + a_1(t)y' + a_0(t)y = f(t). \quad (2.8)$$

Note that this is not necessarily a constant coefficient equation⁷. Let y_1 and y_2 be solutions to the associated homogeneous differential equation. Then we know the solution to equation 2.8 will have the form

$$y = y_p + \alpha y_1 + \beta y_2.$$

⁷ This is not particularly useful right now since we can only solve homogeneous constant coefficient equations, but once we've spoken about Cauchy-Euler equations this becomes relevant.

Variation of parameters is a method for determining the particular solution y_p from the complementary solution $\alpha y_1 + \beta y_2$. The basic idea behind variation of parameters is to assume⁸ that the particular solution can be written in the form

$$y_p = u_1 y_1 + u_2 y_2 \quad (2.9)$$

where u_1 and u_2 are some functions that we will have to determine. At this point we have two unknown functions but only one equation (2.8) so we make a second assumption:

$$u_1' y_1 + u_2' y_2 = 0. \quad (2.10)$$

This looks odd, but it makes our lives much easier, since if we know differentiate the expression for y_p so that we can substitute back into the original equation we find

$$\begin{aligned} y_p' &= u_1' y_1 + u_1 y_1' + u_2' y_2 + u_2 y_2' \\ &= u_1 y_1' + u_2 y_2' \end{aligned}$$

and

$$y_p'' = u_1' y_1' + u_1 y_1'' + u_2' y_2' + u_2 y_2''.$$

Substitution into equation 2.8 gives us

$$a_2(u_1' y_1' + u_1 y_1'' + u_2' y_2' + u_2 y_2'') + a_1(u_1 y_1' + u_2 y_2') + a_0(u_1 y_1 + u_2 y_2) = f.$$

After rearranging some terms we have

$$u_1(a_2 y_1'' + a_1 y_1' + a_0 y_1) + u_2(a_2 y_2'' + a_1 y_2' + a_0 y_2) + a_2 u_1' y_1' + a_2 u_2' y_2' = f.$$

The first two terms are zero since y_1 and y_2 solve the homogeneous differential equation. What remains can be rewritten as

$$y_1' u_1' + y_2' u_2' = \frac{f}{a_2}. \quad (2.11)$$

The two equations 2.10 and 2.11 can be written more compactly as a matrix equation

$$\underbrace{\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix}}_M \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{f}{a_2} \end{pmatrix}. \quad (2.12)$$

The matrix M in the equation above should look familiar – it's the Wronskian matrix. Since y_1 and y_2 are linearly independent solutions to the same linear differential equation their Wronskian is nonzero (Theorem 1.3) and the matrix M is invertible. We can therefore write

$$\begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = M^{-1} \begin{pmatrix} 0 \\ \frac{f}{a_2} \end{pmatrix}$$

⁸ This assumption was made in the late 1700's by Euler and Lagrange. They were trying to work out orbits of various celestial bodies. The justification for this assumption is that if the function f is 'small' then the particular solution should look similar to the complementary solution. It turns out this works even if the function f isn't 'small', whatever that means...

which gives expressions for u'_1 and u'_2 . Integrate these to recover u_1 and u_2 and substitute into equation 2.9 to find y_p .

We're not going to go through this entire process every time we solve a differential equation. We will pick up at 2.12 instead. Make sure that you know how to derive this equation.

EXAMPLE. Solve $y'' - y = e^t$ by using variation of parameters.

Before we start there is a very important comment to be made about this problem, and problems like it: it can be solved by the annihilator method, it would be much easier to solve it by the annihilator method, and we should therefore (if not explicitly told otherwise) solve it using the annihilator method. Of course you're welcome to use variation of parameters if you prefer⁹.

First we solve $y'' - y = 0$ and find $y_1 = e^t$ and $y_2 = e^{-t}$. Assume that $y_p = u_1 e^t + u_2 e^{-t}$ and pick up from 2.12. In this case it's

$$\begin{pmatrix} e^t & e^{-t} \\ e^t & -e^{-t} \end{pmatrix} \begin{pmatrix} u'_1 \\ u'_2 \end{pmatrix} = \begin{pmatrix} 0 \\ e^t \end{pmatrix}.$$

It's easy to invert a 2×2 matrix¹⁰ and after multiplying through by the inverse we have

$$\begin{pmatrix} u'_1 \\ u'_2 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} -e^{-t} & -e^{-t} \\ -e^t & e^t \end{pmatrix} \begin{pmatrix} 0 \\ e^t \end{pmatrix}$$

which gives $u'_1 = \frac{1}{2}$ and $u'_2 = -\frac{1}{2}e^{2t}$ once we multiply out. Integrate to recover u_1 and u_2

$$\begin{aligned} u_1 &= \int \frac{1}{2} dt = \frac{1}{2}t + A, \\ u_2 &= \int -\frac{1}{2}e^{2t} dt = -\frac{1}{4}e^{2t} + B. \end{aligned}$$

We shall see shortly that the constants of integration A and B are not necessary but we keep them in place for the moment. The particular solution is

$$\begin{aligned} y_p &= u_1 y_1 + u_2 y_2 \\ &= e^t \left(\frac{1}{2}t + A \right) + e^{-t} \left(-\frac{1}{4}e^{2t} + B \right) \\ &= \frac{1}{2}te^t + Ae^t - \frac{1}{4}e^t + Be^{-t} \end{aligned}$$

and the general solution is

$$y = \frac{1}{2}te^t + Ae^t - \frac{1}{4}e^t + Be^{-t} + \alpha e^t + \beta e^{-t}.$$

Note that there is some duplication of terms in the particular and complementary solutions. The terms that arise from the constants

⁹ It's like taking an Uber from Smuts to Fuller. It works, but people will think you're a bit odd.

¹⁰ $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

of integration will always be subsumed by the complementary solution. In this case there is an additional term, $-\frac{1}{4}e^t$, that can also be absorbed into the complementary solution. Our final, tidier answer is

$$y = \frac{1}{2}te^t + \alpha e^t + \beta e^{-t}.$$

EXAMPLE. Solve the differential equation $y'' + 9y = \tan(3t)$.

It's not specified how we should solve this problem, but after some time unsuccessfully trying to find an annihilator for $\tan(3t)$ you should realise that we have to use variation of parameters.

Again, we first solve the homogeneous equation $y'' + 9y = 0$ and recover the two independent solutions $y_1 = \cos(3t)$ and $y_2 = \sin(3t)$. Now assume $y_p = u_1 \cos(3t) + u_2 \sin(3t)$ and pick up from 2.12.

$$\begin{pmatrix} \cos(3t) & \sin(3t) \\ -3\sin(3t) & 3\cos(3t) \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ \tan(3t) \end{pmatrix}.$$

After inverting we have

$$\begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 3\cos(3t) & -\sin(3t) \\ 3\sin(3t) & \cos(3t) \end{pmatrix} \begin{pmatrix} 0 \\ \tan(3t) \end{pmatrix}$$

and multiplication gives $u_1' = -\frac{1}{3}\sin(3t)\tan(3t)$ and $u_2' = \frac{1}{3}\sin(3t)$.

These two expressions must be integrated

$$\begin{aligned} u_1 &= \int -\frac{1}{3}\sin(3t)\tan(3t)dt, \\ &= -\frac{1}{3} \int \frac{1 - \cos^2(3t)}{\cos(3t)} dt, \\ &= \frac{1}{9}\sin(3t) - \frac{1}{3}\ln(\sec(3t) + \tan(3t)), u_2 = \int \frac{1}{3}\sin(3t)dt = -\frac{1}{9}\cos(3t). \end{aligned}$$

The particular solution is

$$\begin{aligned} y_p &= \frac{1}{9}\sin(3t)\cos(3t) - \frac{1}{3}\cos(3t)\ln(\sec(3t) + \tan(3t)) - \frac{1}{9}\cos(3t)\sin(3t) \\ &= -\frac{1}{3}\cos(3t)\ln(\sec(3t) + \tan(3t)) \end{aligned}$$

and the general solution is

$$y = -\frac{1}{3}\cos(3t)\ln(\sec(3t) + \tan(3t)) + \alpha \cos(3t) + \beta \sin(3t).$$

Variation of parameters can also be used to solve nonhomogeneous n -th order equations. Consider

$$a_n(t)y^{(n)} + a_{n-1}(t)y^{(n-1)} + \dots + a_1(t)y' + a_0(t)y = f(t).$$

The general solution has the form $y = y_p + y_0$ where y_0 is the complementary solution $\alpha_1 y_1 + \alpha_2 y_2 + \dots + \alpha_n y_n$ that solves the associated homogeneous equation. We assume that the particular solution takes the form

$$y_p = u_1 y_1 + u_2 y_2 + \dots + u_n y_n.$$

Then

$$y'_p = u'_1 y_1 + u_1 y'_1 + u'_2 y_2 + u_2 y'_2 + \dots + u'_n y_n + u_n y'_n.$$

To simplify this expression we assume that the sum

$$u'_1 y_1 + u'_2 y_2 + \dots + u'_n y_n = 0$$

as we did for the second order case. Then

$$y'_p = u_1 y'_1 + u_2 y'_2 + \dots + u_n y'_n.$$

Now we calculate the next derivative

$$y''_p = u'_1 y'_1 + u_1 y''_1 + u'_2 y'_2 + u_2 y''_2 + \dots + u'_n y'_n + u_n y''_n.$$

Again we simplify this by making the assumption that

$$u'_1 y'_1 + u'_2 y'_2 + \dots + u'_n y'_n = 0$$

so

$$y''_p = u_1 y''_1 + u_2 y''_2 + \dots + u_n y''_n.$$

We continue in this way, taking derivatives, making simplifying assumptions, until we reach the n -th derivative

$$y_p^{(n)} = u'_1 y_1^{(n-1)} + u_1 y_1^{(n)} + u'_2 y_2^{(n-1)} + u_2 y_2^{(n)} + \dots + u'_n y_n^{(n-1)} + u_n y_n^{(n)}.$$

This expression cannot be simplified in the same way because the n functions $u_1, u_2, \dots, u_n$ are already constrained by n equations: the original differential equation and the $n - 1$ assumptions made on the previous derivatives. Substituting the expressions for $y'_p, y''_p, \dots, y_p^{(n)}$ into the original differential equation, rearranging terms and using the fact that $y_1, y_2, \dots, y_n$ solve the homogeneous differential equation to set many of the terms to zero yields

$$y_1^{(n-1)} u'_1 + y_2^{(n-1)} u'_2 + \dots + y_n^{(n-1)} u'_n = \frac{f}{a_n}.$$

This, along with the assumptions we made to simplify the derivatives can be compactly written as

$$\underbrace{\begin{pmatrix} y_1 & y_2 & \cdots & y_n \\ y'_1 & y'_2 & \cdots & y'_n \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{pmatrix}}_M \begin{pmatrix} u'_1 \\ u'_2 \\ \vdots \\ u'_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ \frac{f}{a_n} \end{pmatrix}.$$

Again, we recognise that M is the Wronskian matrix which is invertible since $\{y_1, y_2, \dots, y_n\}$ is linearly independent. We can therefore solve for u'_i , integrate to recover u_i and produce y_p . This becomes very difficult to do by hand for $n > 3$, and even $n = 3$ is not easy.

EXAMPLE. Solve the differential equation $D(D - I)(D + I)y = t$ by variation of parameters.

Note, as before, that this problem can be solved by the annihilator method and if we weren't illustrating variation of parameters that is what we would do.

The solutions to the associated homogeneous equation are $y_1 = 1$, $y_2 = e^t$ and $y_3 = e^{-t}$. The particular solution therefore has the form

$$y_p = u_1 + e^t u_2 + e^{-t} u_3.$$

To find these unknown functions u_1 to u_3 we use variation of parameters. Skipping the boring details we pick up from

$$\begin{pmatrix} 1 & e^t & e^{-t} \\ 0 & e^t & -e^{-t} \\ 0 & e^t & e^{-t} \end{pmatrix} \begin{pmatrix} u'_1 \\ u'_2 \\ u'_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ t \end{pmatrix}.$$

We can either invert this matrix and multiply through to find u'_1 , u'_2 or u'_3 , or (a tiny bit faster) we can Gauss reduce the augmented matrix

$$\left(\begin{array}{ccc|c} 1 & e^t & e^{-t} & 0 \\ 0 & e^t & -e^{-t} & 0 \\ 0 & e^t & e^{-t} & t \end{array} \right).$$

Either way we should find that

$$\begin{aligned} u_1 &= -\int t dt = -\frac{t^2}{2}, \\ u_2 &= \int \frac{1}{2} t e^{-t} dt = -\frac{1}{2}(t+1)e^{-t}, \\ u_3 &= \int \frac{1}{2} t e^t dt = \frac{1}{2}(t-1)e^t. \end{aligned}$$

The particular solution is therefore

$$\begin{aligned} y_p &= -\frac{t^2}{2} - \frac{1}{2}(t+1) + \frac{1}{2}(t-1) \\ &= -\frac{t^2}{2} - 1 \end{aligned}$$

and the general solution is

$$y = -\frac{t^2}{2} + \alpha_1 + \alpha_2 e^t + \alpha_3 e^{-t}.$$

Cauchy-Euler Equations

Until this point we've restricted ourselves to solving differential equations with constant coefficients. We've done this because they're easy to solve. All we need to do is to find an auxiliary polynomial, factorise it, and the solution is then easily recovered from the roots of that polynomial.

Unfortunately many of the most interesting differential equations do not have constant coefficients. For example, Bessel's equation

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - \alpha^2)y = 0$$

describes the modes of vibration of a circular acoustic membrane like the skin of a drum. It doesn't have solutions in terms of elementary functions. Its solutions are the Bessel functions, denoted $J_\alpha(x)$ and $Y_\alpha(x)$, which are defined as those functions that solve Bessel's equation.

Similarly, the Airy function $\text{Ai}(x)$, is defined as the solution to the Airy equation

$$\frac{d^2 y}{dx^2} - xy = 0$$

and it's important in optics – it describes the intensity of a rainbow.

We could learn to work with these special functions in the same way that we had to learn to work with trigonometric and exponential functions at school. We're not going to. Instead we are going to tackle one class of linear differential equations without constant coefficients that does have solutions in terms of the elementary functions we're used to.

DEFINITION. A homogeneous **Cauchy-Euler equation** is a linear differential equation that has the form

$$x^n \frac{d^n y}{dx^n} + x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + x \frac{dy}{dx} + y = 0.$$

Note that we are using x rather than t as the independent variable in this equation. The reason for that will become shortly apparent.

THEOREM 2.8. A Cauchy-Euler differential equation can be reduced to a constant coefficient differential equation by making the change of variable $t = \ln(x)$.

Proof. This is a tutorial problem. □

REMARK. A similar change of variable is important later in the course when we solve the Black-Scholes equation.

In the absence of a proof let's proceed directly to some examples to get a sense for how this works. In what follows we will make frequent use of the chain rule

$$\frac{d}{dx} [f(g(x))] = \frac{df}{dg} \frac{dg}{dx}.$$

EXAMPLE. Solve the differential equation $xy' + 2y = 0$.¹¹

Let $t = \ln(x)$. Then

$$\begin{aligned} y' &= \frac{dy}{dx} \\ &= \frac{dy}{dt} \frac{dt}{dx} \\ &= \frac{1}{x} \frac{dy}{dt} \\ &= \frac{1}{x} \dot{y}. \end{aligned}$$

¹¹ We use y' for the derivative of y with respect to x and $\dot{y}$ for the derivative of y with respect to t .

Substituting in gives

$$\dot{y} + 2y = 0.$$

The solution is therefore

$$\begin{aligned} y &= \alpha e^{-2t} \\ &= \alpha e^{-2\ln(x)} \\ &= \frac{\alpha}{x^2} \end{aligned}$$

Things become a little trickier for second-order equations.

EXAMPLE. Solve the differential equation $x^2 y'' - 3xy' + 3y = 0$.

Let $t = \ln(x)$. Then $y' = \frac{1}{x} \dot{y}$ as we showed above and

$$\begin{aligned} y'' &= \frac{d}{dx} \left[\frac{1}{x} \dot{y} \right] \\ &= \frac{d}{dx} \left[\frac{1}{x} \right] \dot{y} + \frac{1}{x} \frac{d}{dx} [\dot{y}] \\ &= -\frac{1}{x^2} \dot{y} + \frac{1}{x^2} \ddot{y}. \end{aligned}$$

Substitute in to the differential equation and we find that

$$\ddot{y} - \dot{y} - 3\dot{y} + 3y = 0$$

$$\ddot{y} - 4\dot{y} + 3y = 0.$$

The auxiliary polynomial is $p(\lambda) = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3)$. The solution is therefore

$$\begin{aligned} y &= \alpha e^t + \beta e^{3t} \\ &= \alpha e^{\ln(x)} + \beta e^{3\ln(x)} \\ &= \alpha x + \beta x^3. \end{aligned}$$

REMARK. In both of these examples the constant coefficient differential equation had an auxiliary polynomial with distinct real roots. If we have repeated roots, or a complex conjugate pair of roots, we end up with more complicated solutions, e.g. $y = \alpha te^{2t}$ becomes $y = \alpha \ln(x)x^2$.

REMARK. Both of these examples were homogeneous differential equations. If the equation is nonhomogeneous you can either make the same change of variable to the nonhomogeneous term, solve using the annihilator method or variation of parameters (working in t), and then change everything, including the particular solution, back. Or, you can solve the associated homogeneous equation and then use variation of parameters (working in x) to find the particular solution.

3

Systems of Linear Ordinary Differential Equations

In the previous chapter we tackled ordinary differential equations. In this chapter we will deal with systems of ordinary differential equations.

DEFINITION. A **system of linear differential equations** is a system of equations involving multiple unknown functions $x_1, x_2, \dots, x_n$ and their first derivatives that can be written in the following way

$$\begin{aligned}\frac{dx_1}{dt} &= a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \\ \frac{dx_2}{dt} &= a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \\ &\vdots \\ \frac{dx_n}{dt} &= a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n.\end{aligned}$$

In principle the coefficients a_{ij} could be functions of t . We will restrict ourselves to the case of constant coefficients which is reasonably straightforward to solve. In this case the system can be rewritten more compactly as

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x}$$

where $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$ and the entries in the constant matrix A are the coefficients a_{ij} . One of the reasons we're interested in systems of differential equations is because every n -th order differential equation can be rewritten as a system of n first-order equations. Given the n -th order differential equation

$$a_n(t) \frac{d^n y}{dt^n} + a_{n-1}(t) \frac{d^{n-1} y}{dt^{n-1}} + \dots + a_0(t)y = f(t)$$

we introduce new variables $x_{j+1} = \frac{d^j y}{dt^j}$. Then

$$\begin{aligned}\frac{dx_1}{dt} &= x_2 \\ \frac{dx_2}{dt} &= x_3 \\ &\vdots \\ \frac{dx_n}{dt} &= -\frac{1}{a_n(t)} \left(a_{n-1}(t) \frac{d^{n-1}y}{dt^{n-1}} + \dots + a_0(t)y - f(t) \right),\end{aligned}$$

or

$$\frac{d\mathbf{x}}{dt} = A(t)\mathbf{x} + \mathbf{f}(t).$$

This is a useful thing to do because in the non-constant coefficient case there isn't a neat analytic solution and so we'd have to generate a numerical solution. Computers are much better at dealing with a big first-order system than a single n -th order equation. It's also theoretically useful. We shall see that there is a much quicker way to prove Theorem 2.6 that avoids Lemma 2.5.

One consequence of this observation is that we should expect a lot of this chapter to look familiar. We must be able to recover all the results from the previous chapter by looking at special cases in this chapter.

Diagonalisable Systems

Consider a homogeneous system of n first-order differential equations with constant coefficients

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x}.$$

If the matrix A is a diagonal matrix then we say the system is **uncoupled** and it's very easy to solve. It reduces to n first-order differential equations that have no influence on each other.

EXAMPLE. Solve the system of differential equations

$$\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 5 & 0 \\ 0 & -3 \end{pmatrix} \mathbf{x}.$$

You should show¹ that the solution is

¹ Do it.

$$\mathbf{x} = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{5t} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-3t}.$$

Obviously not all examples are this easy.

EXAMPLE. Solve the system of differential equations

$$\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 2 & -1 \\ 5 & -4 \end{pmatrix} \mathbf{x}.$$

The matrix A is not diagonal. It is however the next best thing – it's a diagonalisable matrix. We can see this if we calculate its eigenvalues and eigenvectors. First the eigenvalues:

$$\begin{aligned} A &= \begin{pmatrix} 2 & -1 \\ 5 & -4 \end{pmatrix} \\ |A - \lambda I| &= \begin{vmatrix} 2 - \lambda & -1 \\ 5 & -4 - \lambda \end{vmatrix} \\ &= (2 - \lambda)(-4 - \lambda) + 5 \\ &= \lambda^2 + 2\lambda - 3 \\ &= (\lambda + 3)(\lambda - 1) \end{aligned}$$

The roots of the characteristic polynomial (and hence the eigenvalues of A) are at $\lambda_1 = 1$ and $\lambda_2 = -3$. The corresponding eigenvectors are

$$\begin{aligned} \lambda_2 = 1: \quad & \left(\begin{array}{cc|c} 1 & -1 & 0 \\ 5 & -5 & 0 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array} \right) \\ & \Rightarrow \mathbf{v}_1 = \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (\alpha \neq 0). \\ \lambda_2 = -3: \quad & \left(\begin{array}{cc|c} 5 & -1 & 0 \\ 5 & -1 & 0 \end{array} \right) \sim \left(\begin{array}{cc|c} 5 & -1 & 0 \\ 0 & 0 & 0 \end{array} \right) \\ & \Rightarrow \mathbf{v}_2 = \beta \begin{pmatrix} 1 \\ 5 \end{pmatrix} \quad (\beta \neq 0). \end{aligned}$$

Since there are two linearly independent eigenvectors, and hence there is a basis of eigenvectors for $\mathbb{R}^2$, A is diagonalisable. We introduce the matrix

$$P = \begin{pmatrix} 1 & 1 \\ 1 & 5 \end{pmatrix}$$

whose columns are the eigenvectors. The system of differential equations we want to solve can be decoupled by introducing a two new variables y_1 and y_2 that are appropriate linear combinations of x_1 and x_2 :

$$\mathbf{x} = P\mathbf{y}.$$

Substitution into the system of differential equations gives us

$$\begin{aligned}\frac{d}{dt}[\mathbf{x}] &= A\mathbf{x} \\ \frac{d}{dt}[P\mathbf{y}] &= AP\mathbf{y} \\ P\frac{d}{dt}[\mathbf{y}] &= AP\mathbf{y} \\ \frac{d\mathbf{y}}{dt} &= P^{-1}AP\mathbf{y} \\ \frac{d\mathbf{y}}{dt} &= D\mathbf{y}.\end{aligned}$$

The entries of the diagonal matrix D are simply the eigenvalues of A . This is again easy to solve

$$\mathbf{y} = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-3t}.$$

The final step is to convert back into the original variables using $\mathbf{x} = P\mathbf{y}$.

$$\mathbf{x} = \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + \beta \begin{pmatrix} 1 \\ 5 \end{pmatrix} e^{-3t}.$$

EXAMPLE. Solve the system

$$\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 3 & 2 \\ 1 & 2 & 3 \end{pmatrix} \mathbf{x}.$$

Some calculation shows that the eigenvalues of B are $\lambda_1 = 5$ and $\lambda_{2,3} = 1$ (i.e. this eigenvalue has an algebraic multiplicity of 2). The corresponding eigenvectors are

$$\mathbf{v}_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}.$$

Fortunately the geometric multiplicity of the repeated eigenvalue is also 2 so there is a basis of eigenvectors. Next we introduce a matrix P whose columns are simply these eigenvectors, let $\mathbf{x} = P\mathbf{y}$ which decouples the system, solve, and translate back into the original variables. The result is

$$\mathbf{x} = \alpha \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} e^{5t} + \beta \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} e^t + \gamma \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} e^t.$$

The structure of this problem, and of the resulting solution, is similar to the previous example's. This holds generally.

THEOREM 3.1. If A is an $n \times n$ diagonalisable matrix with real (possibly repeated) eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ and n linearly independent corresponding eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ then the general solution to the system

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x}$$

is

$$\mathbf{x} = \sum_{i=1}^n \alpha_i \mathbf{v}_i e^{\lambda_i t}.$$

The system has a fundamental set

$$\left\{ \mathbf{v}_1 e^{\lambda_1 t}, \mathbf{v}_2 e^{\lambda_2 t}, \dots, \mathbf{v}_n e^{\lambda_n t} \right\}$$

and hence the solution space is n -dimensional.

When solving systems of differential equations we often refer to a **fundamental matrix** rather than a fundamental set. This is simply an invertible matrix whose columns are solutions to the differential equation, e.g.

$$\left(\mathbf{v}_1 e^{\lambda_1 t} \mid \mathbf{v}_2 e^{\lambda_2 t} \mid \dots \mid \mathbf{v}_n e^{\lambda_n t} \right)$$

Proof. There are two parts to the proof. First we will show that the claimed expression really is a solution. Let $\mathbf{x} = \sum_{i=1}^n \alpha_i \mathbf{v}_i e^{\lambda_i t}$, then

$$\begin{aligned} \frac{d\mathbf{x}}{dt} &= \frac{d}{dt} \left[\sum_{i=1}^n \alpha_i \mathbf{v}_i e^{\lambda_i t} \right] \\ &= \sum_{i=1}^n \alpha_i \mathbf{v}_i \frac{d}{dt} \left[e^{\lambda_i t} \right] \\ &= \sum_{i=1}^n \alpha_i \lambda_i \mathbf{v}_i e^{\lambda_i t} \\ &= \sum_{i=1}^n \alpha_i A \mathbf{v}_i e^{\lambda_i t} \\ &= A \left[\sum_{i=1}^n \alpha_i \mathbf{v}_i e^{\lambda_i t} \right] \\ &= A\mathbf{x}. \end{aligned}$$

The second part is to show that there are no other solutions. A rigorous proof of this will have to wait until we have discussed the matrix exponential (which is the next section, so it's not a long wait). We might believe it's true since an n -th order differential equation has an n -dimensional solution space and can be converted to a system of n first-order equations, and so we expect this system to have an n -dimensional solution space as well. $\square$

The Matrix Exponential

In the section above we discussed how to solve a system of linear differential equations when the matrix A that represents the system is diagonalisable. Not all matrices are diagonalisable, so the approach above isn't always going to work. In this section we introduce the matrix exponential, which shares many similarities with the regular exponential, and use it to solve diagonalisable systems. Once we know how to solve these easy systems we tackle the problem of solving nondiagonalisable systems with the matrix exponential in the sections below.

The exponential function we met in first year, $\exp : \mathbb{R} \rightarrow \mathbb{R}$, often denoted simply by e^x is defined by

$$\exp(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

We know that this infinite sum converges² for any value of x . The matrix exponential is defined similarly. If A is an $n \times n$ matrix then we define

$$\exp(A) = \sum_{k=0}^{\infty} \frac{A^k}{k!}.$$

Convergence of the matrix exponential is not obvious³.

The matrix exponential has a number of useful properties.

1. If A and B are both $n \times n$ matrices, and they commute (i.e. $AB = BA$) then

$$e^A e^B = e^{A+B} = e^B e^A.$$

The proof of this statement relies on the Cauchy product and the binomial theorem. It's long and not particularly interesting so we won't go through it here, but it is good practice in working carefully with indexes so you could do it if you want.

Note that e^A is always an invertible matrix since A and $-A$ commute. Then $e^A e^{-A} = e^0 = I$.

2. If P is invertible then

$$e^{P^{-1}AP} = P^{-1}e^A P.$$

The proof is relatively straightforward. The main idea is that

$$(P^{-1}AP)^k = (P^{-1}AP)(P^{-1}AP) \cdots (P^{-1}AP) = P^{-1}A^k P.$$

You should try to turn this observation into a formal proof.

² Prove it!

³ Interested students should be able to follow a proof once they've completed 2RA. You can find it online or in any decent textbook.

3. Let $f : t \mapsto e^{tA}$. Then the derivative of f is

$$\frac{d}{dt}f = \frac{d}{dt}[e^{tA}] = Ae^{tA}.$$

To prove this we simply differentiate term-by-term⁴.

$$\begin{aligned} D[e^{tA}] &= \frac{d}{dt} \left[\sum_{k=0}^{\infty} \frac{(tA)^k}{k!} \right] \\ &= \sum_{k=0}^{\infty} \frac{d}{dt} \left[\frac{t^k A^k}{k!} \right] \\ &= \sum_{k=1}^{\infty} \frac{k t^{k-1} A^k}{k!} \\ &= \sum_{k=1}^{\infty} A \frac{t^{k-1} A^{k-1}}{(k-1)!} \\ &= A \sum_{\kappa=0}^{\infty} \frac{t^{\kappa} A^{\kappa}}{\kappa!} \\ &= Ae^{tA} \end{aligned}$$

⁴ Careful! Term-by-term differentiation of an infinite series is, in general, not a good idea. Although we have not shown it the matrix exponential converges uniformly and this doesn't cause any problems.

We can use this to finish the proof of Theorem 3.1 that we started above. In fact, we prove something a little stronger.

THEOREM 3.2. *A linear, constant coefficient system of n first-order differential equations*

$$\frac{dx}{dt} = Ax \tag{3.1}$$

has an n -dimensional solution space. The solution is

$$x = e^{At}c$$

where c is a vector of arbitrary constants.

Note that we are not imposing any conditions on the eigenvalues of A , or requiring that it is diagonalisable.

Proof. The proof follows the same idea as the proof of Theorem 2.1.

Let x be a solution to 3.1. Then

$$\begin{aligned} \frac{d}{dt} [e^{-tA}x] &= \frac{d}{dt} [e^{-tA}]x + e^{-tA} \frac{dx}{dt} \\ &= -Ae^{-tA}x + e^{-tA}Ax \\ &= -Ae^{-tA}x + Ae^{-tA}x \\ &= \mathbf{0} \end{aligned}$$

This implies that $e^{-tA}x$ is constant, so

$$e^{-tA}x = c$$

which in turn implies that

$$\mathbf{x} = e^{tA}\mathbf{c}.$$

The right-hand side of this equation is simply a linear combination of the columns of e^{tA} . Since e^{tA} is invertible its n columns are linearly independent, and each of its columns is a solution to the differential equation, so the solution space is n -dimensional. $\square$

Theorem 3.2 is important. You should be able to use it to prove Theorem 2.6 without Lemma 2.5. Simply convert the n -th order differential equation into a system of n equations and use the matrix exponential⁵.

Theorem 3.2 is also going to allow us to solve nondiagonalisable systems and nonhomogeneous systems. All we need to do is calculate a matrix exponential. When A is diagonalisable this is relatively straightforward (we do an example below), but in general this is a difficult problem. The remaining sections in this chapter cover the details.

⁵ Work through the details yourself. This has been set as an exam problem before.

EXAMPLE. Solve the system of differential equations

$$\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 3 & 2 \\ 1 & 2 & 3 \end{pmatrix} \mathbf{x}.$$

We recognise this as a system we solved in the previous section. We solve it here using the matrix exponential and we expect that we should find the same solution.

As before we start by finding the eigenvalues and eigenvectors of the matrix, then producing a new matrix of eigenvectors P and a diagonal matrix of eigenvalues D such that

$$AP = PD.$$

We know from Theorem 3.2 that the solution has the form

$$\mathbf{x} = e^{tA}\mathbf{c}.$$

To actually calculate the matrix exponential we rewrite A in terms of

P and D .

$$\begin{aligned}
 e^{tA} &= e^{PtDP^{-1}} \\
 &= Pe^{tD}P^{-1} \\
 &= P \left(\sum_{k=0}^{\infty} \frac{(tD)^k}{k!} \right) P^{-1} \\
 &= P \left(\sum_{k=0}^{\infty} \frac{1}{k!} \begin{pmatrix} 5t & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & t \end{pmatrix}^k \right) P^{-1} \\
 &= P \left(\sum_{k=0}^{\infty} \frac{1}{k!} \begin{pmatrix} (5t)^k & 0 & 0 \\ 0 & t^k & 0 \\ 0 & 0 & t^k \end{pmatrix} \right) P^{-1} \\
 &= P \begin{pmatrix} e^{5t} & 0 & 0 \\ 0 & e^t & 0 \\ 0 & 0 & e^t \end{pmatrix} P^{-1}
 \end{aligned}$$

t is a scalar so we can put it wherever we like in the multiplication.

Now the solution can be easily determined

$$\begin{aligned}
 \mathbf{x} &= e^{tA} \mathbf{c} \\
 &= P \begin{pmatrix} e^{5t} & 0 & 0 \\ 0 & e^t & 0 \\ 0 & 0 & e^t \end{pmatrix} P^{-1} \mathbf{c} \\
 &= P \begin{pmatrix} e^{5t} & 0 & 0 \\ 0 & e^t & 0 \\ 0 & 0 & e^t \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} \\
 &= P \begin{pmatrix} \alpha e^{5t} \\ \beta e^t \\ \gamma e^t \end{pmatrix} \\
 &= \alpha \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} e^{5t} + \beta \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} e^t + \gamma \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} e^t
 \end{aligned}$$

This is exactly the same solution that we found earlier.

This method works really easily for diagonalisable systems since the matrix exponential is very easy to calculate. If the system is not diagonalisable the matrix exponential is harder to calculate. We discuss the details below.

Generalised Eigenvectors

Solving diagonalisable systems is easy. We have seen two different ways to do it, either by diagonalising or by using the matrix exponential. It's a little harder when the matrix that describes the system of

ODEs is not diagonalisable. We tackle that problem now, and take the following approach:

1. We solve a diagonalisable system by trial and error, making some sensible guesses as we go based on what we already know about diagonalisable systems.
2. In this process of trial and error we notice a neat way of extending the notion of eigenvectors to what are called generalised eigenvectors.
3. We use the matrix exponential to solve a nondiagonalisable system, using generalised eigenvectors to evaluate the matrix exponential.

EXAMPLE. Solve $\mathbf{x}' = A\mathbf{x}$ where $A = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}$.

The first thing to do is to calculate the eigenvalues and eigenvectors of A .

$$\begin{aligned} p(\lambda) &= |A - \lambda I| \\ &= \begin{vmatrix} 1 - \lambda & -1 \\ 1 & 3 - \lambda \end{vmatrix} \\ &= (1 - \lambda)(3 - \lambda) + 1 \\ &= \lambda^2 - 4\lambda + 4 \\ &= (\lambda - 2)^2 \end{aligned}$$

The matrix A only has one eigenvalue, $\lambda = 2$, with an algebraic multiplicity of 2. What about the eigenvectors? After Gauss reduction

$$\left(\begin{array}{cc|c} 1-2 & -1 & 0 \\ 1 & 3-2 & 0 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

we see that the eigenvectors are

$$\mathbf{v} = \alpha \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \alpha \neq 0$$

so the geometric multiplicity of $\lambda = 2$ is 1. The matrix A is not diagonalisable since we cannot find a basis of eigenvectors. Nonetheless we can still make some reasonable guesses about the solution to this equation. We might try

$$\mathbf{x} = \alpha \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{2t} = \alpha \mathbf{v} e^{2t}$$

and its easy to check that this is a solution since

$$\begin{aligned}\mathbf{x}' &= \alpha \mathbf{v} 2e^{2t} \\ &= 2\alpha \mathbf{v} e^{2t}\end{aligned}$$

and

$$\begin{aligned}A\mathbf{x} &= A(\alpha \mathbf{v} e^{2t}) \\ &= \alpha (A\mathbf{v}) e^{2t} \\ &= 2\alpha \mathbf{v} e^{2t}.\end{aligned}$$

This is a solution, which is a good first step, but it cannot be the entire solution. Since the matrix A is 2×2 we expect a 2-dimensional solution space. There must be another solution. We might try $\mathbf{x} = \beta \mathbf{v} t e^{2t}$, since that sort of thing worked for the repeated roots when we were solving n -th order linear differential equations.

$$\begin{aligned}\mathbf{x}' &= \beta \mathbf{v} (e^{2t} + 2te^{2t}) \\ &= 2\beta \mathbf{v} t e^{2t} + \beta \mathbf{v} e^{2t}\end{aligned}$$

but

$$\begin{aligned}A\mathbf{x} &= A(\beta \mathbf{v} t e^{2t}) \\ &= \beta (A\mathbf{v}) t e^{2t} \\ &= 2\beta \mathbf{v} t e^{2t}.\end{aligned}$$

Unfortunately this doesn't quite work because of the extra term introduced by the product rule. We're quite close though, so maybe we could try $\mathbf{x} = \beta \mathbf{v} t e^{2t} + \gamma \mathbf{w} e^{2t}$ where $\mathbf{w}$ is some vector that has to be determined. Then

$$\begin{aligned}\mathbf{x}' &= \beta \mathbf{v} (e^{2t} + 2te^{2t}) + \gamma \mathbf{w} 2e^{2t} \\ &= 2\beta \mathbf{v} t e^{2t} + \beta \mathbf{v} e^{2t} + 2\gamma \mathbf{w} e^{2t}\end{aligned}$$

and

$$\begin{aligned}A\mathbf{x} &= A(\beta \mathbf{v} t e^{2t}) + A(\gamma \mathbf{w} e^{2t}) \\ &= \beta (A\mathbf{v}) t e^{2t} + \gamma (A\mathbf{w}) e^{2t} \\ &= 2\beta \mathbf{v} t e^{2t} + \gamma (A\mathbf{w}) e^{2t}.\end{aligned}$$

These two expressions agree if

$$\beta \mathbf{v} e^{2t} + 2\gamma \mathbf{w} e^{2t} = \gamma (A\mathbf{w}) e^{2t}$$

which, after some rearrangement, is equivalent to

$$(A - 2I)\mathbf{w} = \mathbf{v}. \quad (3.2)$$

Note that the constants β and γ are not important since the eigenvector $\mathbf{v}$ could be any scalar multiple of $(-1, 1)^T$. In any event we can

solve this

$$\left(\begin{array}{cc|c} -1 & -1 & -1 \\ 1 & 1 & 1 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 0 & 0 \end{array} \right)$$

and we find that

$$\mathbf{w} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \delta \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

The solution to this system will therefore be

$$\mathbf{x} = \alpha \mathbf{v}e^{2t} + \beta(\mathbf{v}te^{2t} + \mathbf{w}e^{2t})$$

or

$$\mathbf{x} = \alpha \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{2t} + \beta \left(t \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) e^{2t}.$$

Note that some part of the vector $\mathbf{w}$ can be subsumed into the first $\alpha \mathbf{v}e^{2t}$ term.

At this stage it is worth pointing out the similarity between equation 3.2 above that we used to calculate $\mathbf{w}$ and $(A - \lambda I)\mathbf{v} = \mathbf{o}$ that is used to calculate eigenvectors of A . This similarity motivates us to introduce and investigate the following:

DEFINITION. If $\mathbf{w}_m$ satisfies

$$(A - \lambda I)^m \mathbf{w}_m = \mathbf{o}$$

but

$$(A - \lambda I)^{m-1} \mathbf{w}_m \neq \mathbf{o}$$

then we call $\mathbf{w}_m$ a **generalised eigenvector of rank m** .

The null space or kernel of $(A - \lambda I)^m$ is the **generalised eigenspace of rank m** . Note that the generalised eigenspace of rank m includes all generalised eigenvectors of rank m or less.

If we have a set of generalised eigenvectors $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k\}$ such that $(A - \lambda I)\mathbf{w}_{p+1} = \mathbf{w}_p$ for all $1 \leq p < k$ and $(A - \lambda I)\mathbf{w}_1 = \mathbf{o}$ then we call that set a **chain of generalised eigenvectors of length k** . Such a chain is a linearly independent set.

REMARK. The regular old eigenvectors that we are used to dealing with are simply generalised eigenvectors of rank 1.

LEMMA 3.3. *Let A be an $n \times n$ matrix, and λ be an eigenvalue of A with an algebraic multiplicity of k . There are k linearly independent generalised eigenvectors corresponding to λ .*

We do not prove this.

LEMMA 3.4. *The intersection of two generalised eigenspaces for different eigenvalues is the zero vector, i.e. if $\lambda_1 \neq \lambda_2$ then*

$$\ker(A - \lambda_1 I)^{k_1} \cap \ker(A - \lambda_2 I)^{k_2} = \{\mathbf{o}\}.$$

Proof. We prove this by induction on the rank k_2 . We start with the base case, i.e. let

$$\mathbf{v} \in \ker(A - \lambda_1 I)^{k_1} \cap \ker(A - \lambda_2 I).$$

Since $\mathbf{v}$ is in the intersection of the kernels, it belongs to each of the kernels. So $\ker(A - \lambda_2 I)\mathbf{v} = \mathbf{0} \Rightarrow A\mathbf{v} = \lambda_2\mathbf{v}$. Then

$$\begin{aligned} (A - \lambda_1 I)^{k_1}\mathbf{v} &= (A - \lambda_1 I)(A - \lambda_1 I) \cdots (A - \lambda_1 I)(A - \lambda_1 I)\mathbf{v} \\ &= (A - \lambda_1 I)(A - \lambda_1 I) \cdots (A - \lambda_1 I)(\lambda_2 - \lambda_1)\mathbf{v} \\ &= (\lambda_2 - \lambda_1)^{k_1}\mathbf{v} \end{aligned}$$

and $(A - \lambda_1 I)^{k_1}\mathbf{v} = \mathbf{0}$ so $(\lambda_2 - \lambda_1)^{k_1}\mathbf{v} = \mathbf{0}$. Since $\lambda_2 \neq \lambda_1$ this is only possible if $\mathbf{v} = \mathbf{0}$ and so the base case is proved.

Assume that $\ker(A - \lambda_1 I)^{k_1} \cap \ker(A - \lambda_2 I)^p = \{\mathbf{0}\}$.

Now consider $\mathbf{v} \in \ker(A - \lambda_1 I)^{k_1} \cap \ker(A - \lambda_2 I)^{p+1}$. Again, if $\mathbf{v}$ is in the intersection it must belong to each kernel separately, so we can say that

$$(A - \lambda_2 I)^{p+1}\mathbf{v} = (A - \lambda_2 I)^p(A - \lambda_2 I)\mathbf{v} = \mathbf{0}.$$

This in turn means that $(A - \lambda_2 I)\mathbf{v}$ belongs to the kernel of $(A - \lambda_2 I)^p$. It also belongs to the kernel of $(A - \lambda_1 I)^{k_1}$ since

$$\begin{aligned} (A - \lambda_1 I)^{k_1}(A - \lambda_2 I)\mathbf{v} &= (A - \lambda_2 I)(A - \lambda_2 I)^{k_1}\mathbf{v} \\ &= \mathbf{0}. \end{aligned}$$

By the inductive assumption we have that $(A - \lambda_2 I)\mathbf{v} = \mathbf{0}$, equivalently $\mathbf{v} \in \ker(A - \lambda_2 I)$, but this is just the base case that we have already proved so $\mathbf{v} = \mathbf{0}$.

Therefore by the principle of mathematical induction this results holds. □

THEOREM 3.5. *An $n \times n$ matrix has n linearly independent generalised eigenvectors. (Hence $\mathbb{R}^n$ has a basis of generalised eigenvectors).*

Proof. This follows from Lemmas 3.3 and 3.4. □

Calculating generalised eigenvectors, particularly for matrices larger than 2×2 or 3×3 , can be tricky (and even the 3×3 case is not always straightforward). We discuss those problems in the following section. What we will do now is to solve the same nondiagonalisable system above by using a neat interaction between generalised eigenvectors and the matrix exponential.

We start with the following observation. If $\mathbf{w}_m$ is a generalised eigenvector of rank m then

$$\begin{aligned} e^{A-\lambda I} \mathbf{w}_m &= \sum_{k=0}^{\infty} \frac{(A-\lambda I)^k}{k!} \mathbf{w}_m \\ &= I\mathbf{w}_m + \frac{1}{2}(A-\lambda I)^2 \mathbf{w}_m + \dots + \frac{1}{(m-1)!} (A-\lambda I)^{m-1} \mathbf{w}_m + \underbrace{\frac{1}{m!} (A-\lambda I)^m \mathbf{w}_m + \dots}_{\text{zero}} \\ &= \sum_{k=0}^{m-1} \frac{(A-\lambda I)^k}{k!} \mathbf{w}_m. \end{aligned}$$

Note that the infinite sum that defines the matrix exponential is truncated to a finite sum when it acts on a generalised eigenvector. Given a differential equation

$$\mathbf{x}' = A\mathbf{x}$$

we know from Theorem 3.2 that its solution is

$$\mathbf{x} = e^{At} \mathbf{c}.$$

The only tricky part is actually evaluating the matrix exponential. To do this we express the constant vector $\mathbf{c}$ as a linear combination of generalised eigenvectors of A and then manipulate the expression to use the truncation we discussed above.

$$\begin{aligned} \mathbf{x} &= e^{At} \mathbf{c} \\ &= \sum_{i=1}^n e^{At} \alpha_i \mathbf{v}_i \\ &= \sum_{i=1}^n e^{At - \lambda_i I t + \lambda_i I t} \alpha_i \mathbf{v}_i \\ &= \sum_{i=1}^n \alpha_i e^{\lambda_i I t} e^{(A - \lambda_i I)t} \mathbf{v}_i \\ &= \sum_{i=1}^n \alpha_i (I e^{\lambda_i t}) \sum_{k=0}^{\infty} \frac{t^k (A - \lambda_i I)^k}{k!} \mathbf{v}_i \\ &= \sum_{i=1}^n \alpha_i e^{\lambda_i t} \sum_{k=0}^{m_i} \frac{t^k (A - \lambda_i I)^k}{k!} \mathbf{v}_i. \end{aligned}$$

The vectors $\mathbf{v}_i$ are generalised eigenvectors, each corresponding to a (possibly repeated) eigenvalue λ_i and with a rank m_i .

This looks like a bit of a mess, but it is a *finite* sum, and as long as we aren't dealing with particularly large matrices, it's not so hard to actually evaluate it.

EXAMPLE. Solve $\mathbf{x}' = A\mathbf{x}$ where $A = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}$.

We established earlier that there is only one eigenvalue, $\lambda = 2$, and a single linearly independent eigenvector

$$\mathbf{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

In the two-dimensional case it's easy to find a second linearly independent generalised eigenvector, we simply look for a chain that terminates on $\mathbf{v}$, i.e. we solve $(A - 2I)\mathbf{w} = \mathbf{v}$, and if we do this we find a second linearly independent generalised eigenvector

$$\mathbf{w} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Our solution is therefore

$$\begin{aligned} \mathbf{x} &= e^{At} \mathbf{c} \\ &= e^{At} (\alpha \mathbf{v} + \beta \mathbf{w}) \\ &= e^{2t} e^{(A-2I)t} (\alpha \mathbf{v} + \beta \mathbf{w}) \\ &= \underbrace{\alpha e^{2t} \mathbf{v} + \beta e^{2t} \left(t^0 (A-2I)^0 \mathbf{w} + t (A-2I) \mathbf{w} \right)}_{i=2} \\ &= \alpha e^{2t} \mathbf{v} + \beta e^{2t} (\mathbf{w} + t \mathbf{v}) \\ &= \alpha \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{2t} + \beta \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right) e^{2t}. \end{aligned}$$

This is conceptually much more elegant than the trial and error solution we produced above. The solution to a linear differential equation is just an exponential, the same as it always is. Unfortunately it's not quite as neat as we would like – it relies on what you might think of as a bit of a cheap trick to evaluate the matrix exponential. We can do better.

Jordan Normal Forms

An $n \times n$ matrix A is diagonalisable if it is similar to a diagonal matrix, i.e. if we can find an invertible P and diagonal D such that $A = PDP^{-1}$. Not all matrices are diagonalisable, but all matrices are similar to a 'nearly diagonal' matrix. 'Nearly diagonal' is not a well-defined term

DEFINITION. The **Jordan block matrix** $J_{\lambda,k}$ is a $k \times k$ matrix with diagonal entries all equal to λ and superdiagonal entries equal to 1, i.e.

$$J_{\lambda,k} = \underbrace{\begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{pmatrix}}_{k \text{ columns}}.$$

A **Jordan matrix** is a matrix built out of Jordan block matrices, e.g. the matrix

$$J = \left(\begin{array}{cc|ccc|c} 5 & 1 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 & 2 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 2 \end{array} \right)$$

is built from three Jordan block matrices. The notation⁶ for this example would be $J = J_{5,2} \oplus J_{2,3} \oplus J_{2,1}$. The structure of a Jordan matrix reflects the structure of its generalised eigenspaces, and although not all matrices are diagonalisable, they are all similar to a Jordan matrix. If $A = PJP^{-1}$ we call J the **Jordan normal form** of the matrix A . The Jordan normal form is not unique since the constituent Jordan blocks can be rearranged.

⁶ This is the direct sum of A and B .

$$A \oplus B = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$$

We shall see that the solution to a nondiagonalisable system of differential equations is determined by the Jordan normal form of the matrix of coefficients. The matrix exponential of a Jordan normal matrix is composed of the matrix exponential of each constituent Jordan block, i.e. the matrix above has an exponential

$$e^J = e^{J_{5,2}} \oplus e^{J_{2,3}} \oplus e^{J_{2,1}}.$$

Each of these Jordan blocks is responsible for the appearance of a chain of generalised eigenvectors in the solution to the nondiagonalisable system.

The Jordan block $J_{\lambda,k}$ has a single eigenvalue λ with an algebraic multiplicity of k and a geometric multiplicity of 1. For example, consider $J_{2,3}$:

$$J_{2,3} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

$$|J_{2,3} - \lambda I| = (2 - \lambda)^3$$

and if we calculate eigenvectors we have

$$\left(\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

which has only one free variable and so the eigenvectors are

$$\mathbf{v} = \alpha \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \alpha \neq 0.$$

The generalised eigenvectors for a Jordan block matrix can all be found by looking for a chain of generalised eigenvectors (sometimes called a **Jordan chain**). In this case

$$\begin{aligned}\text{Let } \mathbf{w}_1 &= \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\ (J - 2I)\mathbf{w}_2 &= \mathbf{w}_1 \\ \left(\begin{array}{ccc|c} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) &\Rightarrow \mathbf{w}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \\ (J - 2I)\mathbf{w}_3 &= \mathbf{w}_2 \\ \left(\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right) &\Rightarrow \mathbf{w}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\end{aligned}$$

and we recover the canonical basis for $\mathbb{R}^3$. Note that we cannot find a generalised eigenvector of rank 4, if we try we end up with the system of equations

$$\left(\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

which has no solutions.

The eigenvalues and (generalised) eigenvectors of a Jordan matrix can be easily calculated. Consider the Jordan matrix we introduced earlier

$$J = \left(\begin{array}{cc|ccc|c} 5 & 1 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 2 \end{array} \right).$$

Since it is upper triangular the eigenvalues are the diagonal entries, or perhaps more usefully, its eigenvalues are simply the eigenvalues of the constituent Jordan blocks. In this case we have an eigenvalue $\lambda_1 = 5$ with an algebraic multiplicity of 2 and another eigenvalue $\lambda_2 = 2$ with an algebraic multiplicity of $3 + 1$. Each Jordan block contributes a single linearly independent eigenvector⁷, so the rank 1

⁷ If you can't see this straight away pick up a pencil and do the calculation. It's quick, I promise.

eigenspaces are

$$E_5 = \left\{ \alpha \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} ; \alpha \in \mathbb{R} \right\}$$

and

$$E_2 = \left\{ \beta \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} ; \beta, \gamma \in \mathbb{R} \right\}.$$

The three additional generalised eigenvectors we need to create a basis are also easy to calculate. There will be a generalised eigenvector of rank 2 corresponding to $\lambda_1 = 5$ since that Jordan block was 2×2 and two generalised eigenvectors of rank 2 and 3 corresponding to $\lambda_2 = 2$ and the 3×3 Jordan block.

LEMMA 3.6. *Let A be an $n \times n$ matrix and $\{w_1, w_2, \dots, w_k\}$ be a chain of generalised eigenvectors corresponding to λ . Then*

$$A(w_1 | w_2 | \dots | w_k) = (w_1 | w_2 | \dots | w_k) J_{\lambda, k}.$$

Proof. Since the generalised eigenvectors form a chain we have that $(A - \lambda I)w_{p+1} = w_p$, equivalently $Aw_{p+1} = \lambda w_{p+1} + w_p$. The rest follows from the definition of matrix multiplication. $\square$

This is a terse proof. Spend some time making sure you understand it.

THEOREM 3.7. *Every square matrix is similar to a Jordan matrix, i.e. if A is an $n \times n$ matrix we can find an invertible P and a Jordan matrix J such that*

$$A = PJP^{-1} \text{ or } J = P^{-1}AP.$$

Proof. Let A be an $n \times n$ matrix. By Theorem 3.5 there is a basis B of generalised eigenvectors for $\mathbb{R}^n$.

We want to use Lemma 3.6 but this requires some preliminary work. First group all the generalised eigenvectors by associated eigenvalue, then within each eigenvalue generate a chain from each eigenvector by repeatedly applying $(A - \lambda I)$ until the zero vector is obtained. Retain only the longest linearly independent chain(s) and discard the unnecessary vectors to produce a new basis $\hat{B}$.

Let P be the matrix whose columns are elements of $\hat{B}$, making sure that all the elements from a chain are adjacent columns, and that their rank increases as the column index increases. Since $\hat{B}$ is a basis P is invertible. By Lemma 3.6 we have

$$AP = PJ$$

where J is a Jordan matrix constituted from the individual Jordan blocks generated by multiplication of the form $A(\mathbf{w}_1|\mathbf{w}_2|\cdots|\mathbf{w}_k)$. □

EXAMPLE. Find the Jordan normal form of $A = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}$.

We know that this matrix has only one eigenvalue, $\lambda = 2$, and that this eigenvalue has a geometric multiplicity of 1. We have previously calculated a chain of two generalised eigenvectors. It was

$$\mathbf{w}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad \& \quad \mathbf{w}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

If we let

$$P = \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}$$

then

$$P^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

and it is easy enough to check that

$$\begin{aligned} P^{-1}AP &= \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \end{aligned}$$

Which is exactly what we expected we would get. Although we've done this example in some detail we won't normally do all these calculations. Once we have the eigenvalues and chains of generalised eigenvectors we should be able to write down the Jordan normal form immediately.

The Jordan normal form of a matrix is useful because calculating the matrix exponential of a Jordan matrix is relatively easy. We start with a lemmas about 2×2 and 3×3 Jordan blocks.

LEMMA 3.8.

$$(J_{\lambda,2})^k = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}^k = \begin{pmatrix} \lambda^k & \binom{k}{1}\lambda^{k-1} \\ 0 & \lambda^k \end{pmatrix}$$

$$(J_{\lambda,3})^k = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}^k = \begin{pmatrix} \lambda^k & \binom{k}{1}\lambda^{k-1} & \binom{k}{2}\lambda^{k-2} \\ 0 & \lambda^k & \binom{k}{1}\lambda^{k-1} \\ 0 & 0 & \lambda^k \end{pmatrix}$$

Proof. We prove the 2×2 case by induction.

First we consider the base case

$$(J_{\lambda,2})^1 = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} \lambda & \binom{1}{1}\lambda^{1-1} \\ 0 & \lambda \end{pmatrix}.$$

Assume that it is true for $k = n$.

Now

$$\begin{aligned} (J_{\lambda,2})^{n+1} &= (J_{\lambda,2})^n (J_{\lambda,2}) \\ &= \begin{pmatrix} \lambda^n & \binom{n}{1}\lambda^{n-1} \\ 0 & \lambda^n \end{pmatrix} \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} \lambda^n \times \lambda & \lambda^n + \binom{n}{1}\lambda^{n-1} \times \lambda \\ 0 & \lambda^n \times \lambda \end{pmatrix} \\ &= \begin{pmatrix} \lambda^{n+1} & \binom{n+1}{1}\lambda^n \\ 0 & \lambda^{n+1} \end{pmatrix} \end{aligned}$$

so this is true for 2×2 Jordan block matrices by the principal of mathematical induction.

The 3×3 case can also be proved by induction, although it's a little trickier. The base case is true since

$$(J_{\lambda,3})^1 = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix} = \begin{pmatrix} \lambda & \binom{1}{1}\lambda^{1-1} & \binom{1}{2}\lambda^{1-2} \\ 0 & \lambda & \binom{1}{1}\lambda^{1-1} \\ 0 & 0 & \lambda \end{pmatrix}.$$

The coefficient $\binom{1}{2}$ looks strange, but it is conventionally defined as zero. Assume that it's true for $k = n$.

Now

$$\begin{aligned}
 (J_{\lambda,3})^{n+1} &= (J_{\lambda,3})^n (J_{\lambda,3})^1 \\
 &= \begin{pmatrix} \lambda^n & \binom{n}{1}\lambda^{n-1} & \binom{n}{2}\lambda^{n-2} \\ 0 & \lambda^n & \binom{n}{1}\lambda^{n-1} \\ 0 & 0 & \lambda^n \end{pmatrix} \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix} \\
 &= \begin{pmatrix} \lambda^{n+1} & \lambda^n + \binom{n}{1}\lambda^n & \binom{n}{1}\lambda^{n-1} + \binom{n}{2}\lambda^{n-1} \\ 0 & \lambda^{n+1} & \lambda^n + \binom{n}{1}\lambda^n \\ 0 & 0 & \lambda^{n+1} \end{pmatrix} \\
 &= \begin{pmatrix} \lambda^{n+1} & \binom{n+1}{1}\lambda^n & \binom{n}{1}\lambda^{n-1} + \binom{n}{2}\lambda^{n-1} \\ 0 & \lambda^{n+1} & \binom{n+1}{1}\lambda^n \\ 0 & 0 & \lambda^{n+1} \end{pmatrix}.
 \end{aligned}$$

The entry in the top-right corner is a little tricky to evaluate, but if we go back to the definition of the binomial coefficients in terms of factorials we see that

$$\begin{aligned}
 \binom{n}{1} + \binom{n}{2} &= n + \frac{n!}{(n-2)!2!} \\
 &= \frac{n(n-2)!2! + n!}{(n-2)!2!} \\
 &= \frac{(n+1)n(n-2)!}{(n-2)!2!} \times \frac{n-1}{n-1} \\
 &= \frac{(n+1)!}{(n-1)!2!} \\
 &= \binom{n+1}{2}
 \end{aligned}$$

and so the result is also true for 3×3 matrices. $\square$

REMARK. The Lemma above can be extended to 4×4 , 5×5 and larger Jordan block matrices. We don't do it here because working with larger systems of differential equations is (for many reasons) computationally infeasible by hand. The 2×2 and 3×3 cases are all we need.

REMARK. A Jordan matrix is constructed out of Jordan block matrices. The k -th power of that Jordan matrix is constructed out of the corresponding k -th powers of its constituent Jordan block matrices.

This Lemma is useful for evaluating matrix exponentials, and hence for solving systems of differential equations.

LEMMA 3.9 (Some useful matrix exponentials).

$$e^{J_{\lambda,2}t} = \begin{pmatrix} e^{\lambda t} & te^{\lambda t} \\ 0 & e^{\lambda t} \end{pmatrix}$$

$$e^{J_{\lambda,3}t} = \begin{pmatrix} e^{\lambda t} & te^{\lambda t} & \frac{1}{2}t^2e^{\lambda t} \\ 0 & e^{\lambda t} & te^{\lambda t} \\ 0 & 0 & e^{\lambda t} \end{pmatrix}$$

Proof.

$$\begin{aligned} e^{J_{\lambda,2}t} &= \sum_{k=0}^{\infty} \frac{1}{k!} \begin{pmatrix} \lambda^k & \binom{k}{1}\lambda^{k-1} \\ 0 & \lambda^k \end{pmatrix} t^k \\ &= \begin{pmatrix} \sum_{k=0}^{\infty} \frac{\lambda^k t^k}{k!} & \sum_{k=1}^{\infty} \frac{k\lambda^{k-1}t^k}{k!} \\ 0 & \sum_{k=0}^{\infty} \frac{\lambda^k t^k}{k!} \end{pmatrix} \\ &= \begin{pmatrix} e^{\lambda t} & t \sum_{k=1}^{\infty} \frac{\lambda^{k-1}t^{k-1}}{(k-1)!} \\ 0 & e^{\lambda t} \end{pmatrix} \\ &= \begin{pmatrix} e^{\lambda t} & te^{\lambda t} \\ 0 & e^{\lambda t} \end{pmatrix} \end{aligned}$$

Check the 3×3 case yourself. □

EXAMPLE. Solve $\mathbf{x}' = A\mathbf{x}$ where $A = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}$.

The solution to this differential equation is $e^{At}\mathbf{c}$. We have previously calculated its Jordan normal form. Now

$$\begin{aligned} e^{At}\mathbf{c} &= e^{PJP^{-1}t}\mathbf{c} \\ &= Pe^{Jt}P^{-1}\mathbf{c} \\ &= P \begin{pmatrix} e^{2t} & te^{2t} \\ 0 & e^{2t} \end{pmatrix} P^{-1}\mathbf{c} \\ &= \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} e^{2t} & te^{2t} \\ 0 & e^{2t} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \\ &= \alpha \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{2t} + \beta \left(\begin{pmatrix} -1 \\ 1 \end{pmatrix} t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) e^{2t} \end{aligned}$$

EXAMPLE. Solve $\mathbf{x}' = B\mathbf{x}$ where $B = \begin{pmatrix} 0 & -3 & 0 \\ -6 & -3 & 0 \\ 2 & 2 & -6 \end{pmatrix}$.

We start by looking for the eigenvalues of B . A short calculation shows that they are $\lambda = 3$ and $\lambda = -6$ with an algebraic multiplicity of 2. The eigenvector corresponding to $\lambda = 3$ is

$$\mathbf{v}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}.$$

Unfortunately, when we look for the eigenvectors corresponding to $\lambda = -6$ we can only find one linearly independent vector,

$$\mathbf{w}_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

The matrix is not diagonalisable. We have to find a generalised eigenvector and it is

$$\mathbf{w}_2 = \begin{pmatrix} \frac{1}{6} \\ \frac{1}{3} \\ 0 \end{pmatrix}.$$

Now for

$$P = \begin{pmatrix} -1 & 0 & 1 \\ 1 & 0 & 2 \\ 0 & 6 & 0 \end{pmatrix}$$

and

$$J = \begin{pmatrix} 3 & 0 & 0 \\ 0 & -6 & 1 \\ 0 & 0 & -6 \end{pmatrix}$$

we have $AP = PJ$ and P is invertible. Note that when we scale the generalised eigenvector $\mathbf{w}_2$ by 6 to eliminate the fractions we have to scale $\mathbf{w}_1$ by 6 as well since the two vectors satisfy the relationship $(B + 6I)\mathbf{w}_2 = \mathbf{w}_1$. Multiplying only one side of the equation is not allowed. Now we write down the solution.

$$\begin{aligned} \mathbf{x} &= e^{Bt} \mathbf{c} \\ &= e^{PJ P^{-1} t} \mathbf{c} \\ &= P e^{J t} P^{-1} \mathbf{c} \\ &= \begin{pmatrix} -1 & 0 & 1 \\ 1 & 0 & 2 \\ 0 & 6 & 0 \end{pmatrix} \begin{pmatrix} e^{3t} & 0 & 0 \\ 0 & e^{-6t} & t e^{-6t} \\ 0 & 0 & e^{-6t} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} \\ &= \alpha \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} e^{3t} + \beta \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix} e^{-6t} + \gamma \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix} t e^{-6t} + \gamma \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} e^{-6t} \end{aligned}$$

Complex Eigenvalues and Eigenvectors

Just as an n -th order linear differential equation with constant coefficients might have an auxiliary polynomial with a complex conjugate pair of roots, so might a system of linear differential equations have a complex conjugate pair of eigenvalues. And just as the complex conjugate roots of the auxiliary polynomial lead to oscillating solutions, so do complex conjugate eigenvalues. We will tackle these types of

problems now, first in an *ad hoc* manner that, with some difficulty, produces a solution, and then second in a more systematic manner where we incorporate complex conjugate eigenvalues into the framework we have developed that uses Jordan normal forms and the matrix exponential to solve systems of differential equations.

EXAMPLE. Solve the differential equation

$$\mathbf{x}' = \underbrace{\begin{pmatrix} -6 & -1 \\ 5 & -4 \end{pmatrix}}_B \mathbf{x}.$$

The characteristic polynomial associated with the matrix B is $p(\lambda) = \lambda^2 + 10\lambda + 29$. This quadratic has a complex conjugate pair of roots

$$\lambda_{1,2} = -5 \pm 2i.$$

We haven't really discussed what to do with complex eigenvalues, but let's push on as before and look for the eigenvectors and hope that things work out. For $\lambda_1 = -5 + 2i$ we have

$$\left(\begin{array}{cc|c} -1-2i & -1 & 0 \\ 5 & 1-2i & 0 \end{array} \right) \sim \left(\begin{array}{cc|c} 1+2i & 1 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

and the eigenvectors here are

$$\mathbf{v}_1 = \alpha \begin{pmatrix} 1 \\ -1-2i \end{pmatrix}.$$

The complex conjugate eigenvalue $\lambda_2 = \overline{\lambda_1}$ has eigenvectors $\mathbf{v}_2 = \overline{\mathbf{v}_1}$. We might write down our solution as

$$\mathbf{x} = \alpha \begin{pmatrix} 1 \\ -1-2i \end{pmatrix} e^{(-5+2i)t} + \beta \begin{pmatrix} 1 \\ -1+2i \end{pmatrix} e^{(-5-2i)t}.$$

This is very unsatisfactory. Our problem was real, our solution should be real. If we choose α and β to be conjugates the imaginary parts of the solution will cancel, and by using Euler's formula to express the complex exponential in terms of sines and cosines we can

make the final expression much more palatable.

$$\mathbf{x} = \left((\Re(\alpha) + i\Im(\alpha)) \begin{pmatrix} 1 \\ -1 - 2i \end{pmatrix} (\cos(2t) + i\sin(2t)) + (\Re(\alpha) - i\Im(\alpha)) \begin{pmatrix} 1 \\ -1 + 2i \end{pmatrix} (\cos(2t) - i\sin(2t)) \right) e^{-5t}$$

which after some long but boring manipulation is

$$\mathbf{x} = \begin{pmatrix} 2\Re(\alpha) \cos(2t) - 2\Im(\alpha) \sin(2t) \\ (-2\Re(\alpha) + 4\Im(\alpha)) \cos(2t) + (4\Re(\alpha) + 2\Im(\alpha)) \sin(2t) \end{pmatrix} e^{-5t}$$

Let $2\Re(\alpha) = a$ and $2\Im(\alpha) = b$, then

$$\begin{aligned} \mathbf{x} = & a \left(\begin{pmatrix} 1 \\ -1 \end{pmatrix} \cos(2t) + \begin{pmatrix} 0 \\ 2 \end{pmatrix} \sin(2t) \right) e^{-5t} \\ & + b \left(\begin{pmatrix} 0 \\ -2 \end{pmatrix} \cos(2t) + \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-5t} \sin(2t) \right) e^{-5t} \end{aligned}$$

Although we haven't shown it this holds more generally. If the matrix A has a complex conjugate pair of eigenvalues λ_1 and λ_2 with corresponding complex conjugate eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ the solution to $\mathbf{x}' = A\mathbf{x}$ will include the terms

$$\begin{aligned} & a(\Re(\mathbf{v}_1) \cos(\Im(\lambda_1)t) - \Im(\mathbf{v}_1) \sin(\Im(\lambda_1)t)) e^{\Re(\lambda_1)t} \\ & + b(\Re(\mathbf{v}_1) \sin(\Im(\lambda_1)t) + \Im(\mathbf{v}_1) \cos(\Im(\lambda_1)t)) e^{\Re(\lambda_1)t}. \end{aligned}$$

This result can be recovered by thinking about the Jordan normal form of a matrix with a complex conjugate pair of eigenvalues and how we might evaluate the matrix exponential in this case. Let A be a matrix that has a complex eigenvalue $\lambda_1 = \alpha + i\beta$ ($\beta \neq 0$) with corresponding eigenvector $\mathbf{u} - i\mathbf{v}$ (There is a choice of sign to be made here; we make this choice so that a matrix we produce later looks familiar). The eigenvalue $\lambda_2 = \overline{\lambda_1}$ will have an eigenvector $\mathbf{u} + i\mathbf{v}$.

LEMMA 3.10. *The matrix A acts on the real and imaginary components of the eigenvectors in the following way:*

$$\begin{aligned} A\mathbf{u} &= \alpha\mathbf{u} + \beta\mathbf{v} \\ A\mathbf{v} &= -\beta\mathbf{u} + \alpha\mathbf{v} \end{aligned}$$

Proof. Apply A to the eigenvector $\mathbf{u} - i\mathbf{v}$.

$$\begin{aligned} A(\mathbf{u} - i\mathbf{v}) &= (\alpha + i\beta)(\mathbf{u} - i\mathbf{v}) \\ &= (\alpha\mathbf{u} + \beta\mathbf{v}) + i(\beta\mathbf{u} - \alpha\mathbf{v}). \end{aligned}$$

By comparing the real and imaginary parts on the left and right side of the equation we see that

$$\begin{aligned} A\mathbf{u} &= \alpha\mathbf{u} + \beta\mathbf{v} \\ A\mathbf{v} &= -\beta\mathbf{u} + \alpha\mathbf{v} \end{aligned}$$

□

LEMMA 3.11. *The set $\{\mathbf{u}, \mathbf{v}\}$ is linearly independent.*

Proof. Consider the linear relation

$$c\mathbf{u} + d\mathbf{v} = \mathbf{0}.$$

We apply the matrix A , using the Lemma above, and rearrange terms.

$$\begin{aligned} c\alpha\mathbf{u} + c\beta\mathbf{v} - d\beta\mathbf{u} + d\alpha\mathbf{v} &= \mathbf{0} \\ \alpha(c\mathbf{u} + d\mathbf{v}) + \beta(-d\mathbf{u} + c\mathbf{v}) &= \mathbf{0} \\ \beta(-d\mathbf{u} + c\mathbf{v}) &= \mathbf{0} \end{aligned}$$

Since $\beta \neq 0$ we must have $-d\mathbf{u} + c\mathbf{v} = \mathbf{0}$, or $c\mathbf{v} = d\mathbf{u}$. Multiply the original linear relation by c . Then

$$\begin{aligned} c^2\mathbf{u} + dc\mathbf{v} &= \mathbf{0} \\ (c^2 + d^2)\mathbf{u} &= \mathbf{0}. \end{aligned}$$

Similarly we can show that $(c^2 + d^2)\mathbf{v} = \mathbf{0}$. This could be solved by setting $\mathbf{u} = \mathbf{v} = \mathbf{0}$, but that would make the eigenvector $\mathbf{u} - i\mathbf{v} = \mathbf{0}$ which is impossible. This means that $c^2 + d^2 = 0$ which makes $c = d = 0$ and so the set is linearly independent. $\square$

LEMMA 3.12. *If A is a 2×2 matrix then*

$$A(\mathbf{u}|\mathbf{v}) = (\mathbf{u}|\mathbf{v}) \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}.$$

This lemma follows from the definition of matrix multiplication. The matrix on the right side of this equation should be familiar. It's exactly the matrix that is associated⁸ with the complex number $z = \alpha + i\beta = \lambda$.

⁸ Details in a tutorial problem

LEMMA 3.13.

$$\exp \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} = e^\alpha \begin{pmatrix} \cos(\beta) & -\sin(\beta) \\ \sin(\beta) & \cos(\beta) \end{pmatrix}.$$

Proof. Let ϕ be the ring homomorphism between $\mathbb{C}$ and a subset of $\mathbb{R}^{2 \times 2}$ defined by

$$\phi : \alpha + i\beta \mapsto \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}.$$

Then

$$\begin{aligned}
 \exp \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} &= \phi \left(\phi^{-1} \left(\exp \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} \right) \right) \\
 &= \phi \left(\phi^{-1} \left(\sum_{k=0}^{\infty} \frac{1}{k!} \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}^k \right) \right) \\
 &= \phi \left(\sum_{k=0}^{\infty} \frac{(\alpha + i\beta)^k}{k!} \right) \\
 &= \phi \left(e^{\alpha + i\beta} \right) \\
 &= \phi \left(e^{\alpha} (\cos(\beta) + i \sin(\beta)) \right) \\
 &= e^{\alpha} \begin{pmatrix} \cos(\beta) & -\sin(\beta) \\ \sin(\beta) & \cos(\beta) \end{pmatrix}
 \end{aligned}$$

□

This is sufficient to solve the systems of differential equations with complex conjugate eigenvalues. We will use matrices of the form

$$\begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}$$

as the constituent Jordan blocks instead of working with

$$\begin{pmatrix} \lambda & 0 \\ 0 & \bar{\lambda} \end{pmatrix}.$$

EXAMPLE. Solve the differential equation

$$\mathbf{x}' = \underbrace{\begin{pmatrix} -6 & -1 \\ 5 & -4 \end{pmatrix}}_B \mathbf{x}.$$

As we have previously discussed, this matrix has a complex conjugate pair of eigenvalues $\lambda_{1,2} = -5 \pm 2i$. The eigenvector that corresponds to λ_1 is

$$\begin{pmatrix} 1 \\ -1 \end{pmatrix} - i \begin{pmatrix} 0 \\ 2 \end{pmatrix}.$$

This means that

$$\begin{pmatrix} -6 & -1 \\ 5 & -4 \end{pmatrix} \underbrace{\begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix}}_P = \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix} \underbrace{\begin{pmatrix} -5 & -2 \\ 2 & -5 \end{pmatrix}}_J.$$

This is more compactly written $BP = PJ$ and since the real and imaginary parts of the eigenvector are linearly independent the matrix P is invertible so $B = PJP^{-1}$.

The solution we want is simply

$$\begin{aligned}
 e^{Bt}\mathbf{c} &= e^{PJ P^{-1}t}\mathbf{c} \\
 &= P e^{Jt} P^{-1}\mathbf{c} \\
 &= P e^{-5t} \begin{pmatrix} \cos(2t) & -\sin(2t) \\ \sin(2t) & \cos(2t) \end{pmatrix} \hat{\mathbf{c}} \\
 &= P \left(a \begin{pmatrix} \cos(2t) \\ \sin(2t) \end{pmatrix} + b \begin{pmatrix} -\sin(2t) \\ \cos(2t) \end{pmatrix} \right) e^{-5t} \\
 &= a \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-5t} \cos(2t) + a \begin{pmatrix} 0 \\ 2 \end{pmatrix} e^{-5t} \sin(2t) \\
 &\quad - b \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-5t} \sin(2t) + b \begin{pmatrix} 0 \\ 2 \end{pmatrix} e^{-5t} \cos(2t)
 \end{aligned}$$

which agrees with the solution we previously calculated for this problem.

EXAMPLE. Solve the system of differential equations

$$\mathbf{x}' = \underbrace{\begin{pmatrix} -2 & 16 & 6 & 10 \\ -2 & -2 & 1 & 4 \\ 4 & -20 & -10 & -16 \\ -6 & 4 & 0 & 2 \end{pmatrix}}_A \mathbf{x}.$$

The eigenvalues of this matrix are $\lambda_1 = -4$ (repeated with an algebraic multiplicity of 2) and $\lambda_{2,3} = -2 \pm 6i$.

First the repeated eigenvalue. Some calculation shows that the corresponding eigenvectors are

$$\mathbf{v}_1 = \alpha \begin{pmatrix} 1 \\ 0 \\ -2 \\ 1 \end{pmatrix}.$$

Since we only have one linearly independent eigenvector, we look for a corresponding generalised eigenvector. In this case we find

$$\mathbf{w}_2 = \frac{\alpha}{4} \begin{pmatrix} 0 \\ 1 \\ -2 \\ 0 \end{pmatrix}.$$

For ease of calculation we choose $\alpha = 4$ to avoid fractions. Then the first two columns of the similarity matrix that will transform between

A and its Jordan normal form are

$$P = \begin{pmatrix} 4 & 0 & \cdot & \cdot \\ 0 & 1 & \cdot & \cdot \\ -8 & -2 & \cdot & \cdot \\ 4 & 0 & \cdot & \cdot \end{pmatrix}.$$

Next we deal with the complex conjugate eigenvalue. Again, some calculation shows that a corresponding eigenvector is

$$\begin{pmatrix} 2 \\ 1 \\ -4 \\ 2 \end{pmatrix} - i \begin{pmatrix} 2 \\ 0 \\ -2 \\ 0 \end{pmatrix}.$$

The final two columns of P are thus determined:

$$P = \begin{pmatrix} 4 & 0 & 2 & 2 \\ 0 & 1 & 1 & 0 \\ -8 & -2 & -4 & -2 \\ 4 & 0 & 2 & 0 \end{pmatrix}$$

and the corresponding Jordan matrix J is

$$J = \begin{pmatrix} -4 & 1 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & -2 & -6 \\ 0 & 0 & 6 & -2 \end{pmatrix}.$$

The solution we are looking for is

$$\begin{aligned} e^{At}\mathbf{c} &= e^{PJ P^{-1}t}\mathbf{c} \\ &= P e^{Jt} P^{-1}\mathbf{c} \\ &= P \begin{pmatrix} e^{-4t} & t e^{-4t} & 0 & 0 \\ 0 & e^{-4t} & 0 & 0 \\ 0 & 0 & e^{-2t} \cos(6t) & -e^{-2t} \sin(6t) \\ 0 & 0 & e^{-2t} \sin(6t) & e^{-2t} \cos(6t) \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \\ &= a \begin{pmatrix} 4 \\ 0 \\ -8 \\ 4 \end{pmatrix} e^{-4t} + b \left(\begin{pmatrix} 0 \\ 1 \\ -2 \\ 0 \end{pmatrix} + \begin{pmatrix} 4 \\ 0 \\ -8 \\ 4 \end{pmatrix} t \right) e^{-4t} + c \left(\begin{pmatrix} 2 \\ 1 \\ -4 \\ 2 \end{pmatrix} \cos(6t) + \begin{pmatrix} 2 \\ 0 \\ -2 \\ 0 \end{pmatrix} \sin(6t) \right) e^{-2t} \\ &\quad + d \left(\begin{pmatrix} 2 \\ 0 \\ -2 \\ 0 \end{pmatrix} \cos(6t) - \begin{pmatrix} 2 \\ 1 \\ -4 \\ 2 \end{pmatrix} \sin(6t) \right) e^{-2t} \end{aligned}$$

Nonhomogeneous Systems

Once we understand Jordan normal forms and the matrix exponential it is conceptually very easy to solve nonhomogeneous systems of differential equations (although there are still some computations that require a little work). The approach should feel familiar; it's analogous to the method of integrating factors used to solve linear first-order equations in MAM1000W. Consider the nonhomogeneous differential equation

$$\mathbf{x}' = A\mathbf{x} + \mathbf{b}.$$

Rearrange and multiply by e^{-At}

$$e^{-At}\mathbf{x}' - Ae^{-At}\mathbf{x} = e^{-At}\mathbf{b}.$$

Recognise that the left-hand side is the result of using the product rule, and integrate on both sides.

$$\begin{aligned} (e^{-At}\mathbf{x})' &= e^{-At}\mathbf{b} \\ e^{-At}\mathbf{x} &= \int e^{-At}\mathbf{b} \, dt \\ \mathbf{x} &= e^{At} \int e^{-At}\mathbf{b} \, dt \\ \mathbf{x} &= Pe^{Jt}P^{-1} \int Pe^{-Jt}P^{-1}\mathbf{b} \, dt \\ \mathbf{x} &= Pe^{Jt} \int e^{-Jt}P^{-1}\mathbf{b} \, dt \end{aligned}$$

The difficulty arises in two places. Firstly, calculating $P^{-1}\mathbf{b}$ is tricky. If we are lucky the nonhomogeneity $\mathbf{b}$ is expressed as a linear combination of the (generalised) eigenvectors of A which means that the resulting multiplication is just the same linear combination of the canonical basis vectors. If we are not lucky we need either to express $\mathbf{b}$ in terms of the eigenvectors ourselves, or invert P . Secondly, the integration may require any of the usual techniques to evaluate (if, indeed, it can even be evaluated in terms of the standard elementary functions).

EXAMPLE. Solve the nonhomogeneous system

$$\mathbf{x}' = \begin{pmatrix} 3 & -2 & -3 \\ 5 & -4 & -3 \\ -1 & 1 & -2 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} t.$$

The first step is to calculate eigenvalues, eigenvectors, generalised eigenvectors and so on. We skip the details (but you can check for yourself). The matrices we will need are

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 0 & 1 & 0 \end{pmatrix}$$

and

$$J = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -2 \end{pmatrix}.$$

Note that the nonhomogeneity is simply t times the first column of P .

The solution is

$$\mathbf{x} = Pe^{Jt} \int e^{-Jt} P^{-1} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} t \, dt$$

$$\mathbf{x} = Pe^{Jt} \int e^{-Jt} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} t \, dt$$

$$\mathbf{x} = Pe^{Jt} \int \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} te^{-t} \, dt$$

$$\mathbf{x} = Pe^{Jt} \left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} (-te^{-t} - e^{-t}) + \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} \right)$$

$$\mathbf{x} = \alpha \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} e^t + \beta \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{-2t} + \gamma \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} e^{-2t} + \gamma \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} te^{-2t} - \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} (t+1)$$

4

Linear Partial Differential Equations

A partial differential equation (PDE) is a differential equation involving an unknown function of two or more variables. For example, the heat equation

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

describes the heat of a one-dimensional object at a time t and a position x . The constant $k > 0$ describes the diffusion in the system and depends on a number of physical factors like the material of the object, its thickness and so on. We restrict ourselves to linear PDEs like the heat equation. What is meant by a linear PDE? If we introduce the notation D_t and D_x for the partial derivatives we can see that the heat equation can be rewritten as

$$(D_t - cD_x^2)u = 0$$

which is just a linear operator acting on the function u (i.e. a linear transformation acting on a vector u). When we discuss the heat equation in more detail we shall see that it behaves exactly as we expect something linear to behave (e.g. it satisfies the principle of superposition).

All PDEs are defined on a particular domain. For the heat equation describing a one-dimensional object of length L for all positive time the domain of $u(x,t)$ is $[0, L] \times [0, \infty)$. On some of the boundaries of the domain the value of the function may be specified. These are the boundary conditions of the problem. At the boundary $t = 0$ we call this boundary condition the initial condition.

The Heat Equation

We start our discussion of the heat equation with the heat equation on a finite domain of length L . In addition to the heat equation

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

we also need some boundary conditions (at $x = 0$ and $x = L$) and an initial condition (at $t = 0$) before we can solve this problem. We consider two different types of boundary condition.

1. **Dirichlet** or **first type** boundary conditions. The value of the unknown function is specified along the boundary of its domain, e.g. if one end of a metal rod was submerged in an ice bath and the other in boiling water we would have $u(0, t) = 0^\circ\text{C}$ and $u(L, t) = 100^\circ\text{C}$.
2. **Neumann** or **second type** boundary conditions. The heat flow, i.e. the derivative of the unknown function u , is specified along the boundary of its domain. If each end of the rod was perfectly insulated and no heat could flow into or out of the rod at either end we would have $u_x(0, t) = u_x(L, t) = 0$.

The boundary conditions are called **homogeneous** if they are satisfied by the zero function and **nonhomogeneous** otherwise. In addition to the two boundary conditions associated with the second spatial derivative there must be an initial condition associated with the time derivative $u(x, t) = u_0(x)$.

Separation of Variables

Initially we restrict ourselves to solving the heat equation in the case of homogeneous Dirichlet boundary conditions:

$$\begin{aligned} \frac{\partial u}{\partial t} &= k \frac{\partial^2 u}{\partial x^2} && \text{for } 0 < x < L \text{ and } 0 < t && \text{PDE} \\ u(0, t) &= 0 && && \text{BC 1} \\ u(L, t) &= 0 && && \text{BC 2} \\ u(x, 0) &= u_0(x) && && \text{IC} \end{aligned}$$

The method of solution of variables makes the assumption that the solution u has a particular form

$$u(x, t) = X(x)T(t).$$

This assumption was made by Bernoulli in the 1700s. It works because it reduces the PDE into two ODEs. If we substitute into the heat equation, take partial derivatives and rearrange we find that

$$\frac{1}{kT} \frac{dT}{dt} = \frac{1}{X} \frac{d^2X}{dx^2}.$$

Note that the left hand side of this equation depends on only time t and the right hand side depends only on position x . The only way these two expressions are equal is if both sides are equal to a real

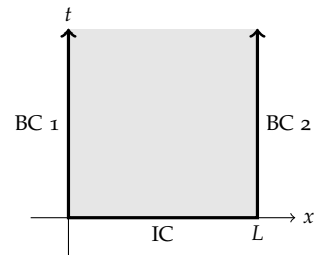


Figure 4.1: Domain of the heat equation

constant, which we call the **separation constant** and denote by C . It will become clear a little later that C must be a negative number, or $C = -\lambda^2$. For now we will proceed working with C and $-\lambda^2$ in parallel. Now we have two *ordinary* differential equations to solve

$$\frac{1}{kT} \frac{dT}{dt} = C (= -\lambda^2) \quad \text{and} \quad \frac{1}{X} \frac{d^2X}{dx^2} = C (= -\lambda^2).$$

We start with the easier time-dependant equation

$$\frac{dT}{dt} = CkT = -\lambda^2 kT.$$

It is clear that the solutions are simply

$$T(t) = Ae^{CkT} = Ae^{-\lambda^2 kT}.$$

Since the diffusion constant k is positive the exponential decays if C is negative, i.e. if $C = -\lambda^2$. This makes physical sense. No matter the initial temperature of the bar we expect that its temperature approaches zero throughout its length as $t \rightarrow \infty$ since the ends of the bars are held fixed at that temperature. We shall see that this physical constraint that $C = -\lambda^2$ appears as a mathematical constraint when we consider the boundary-value problem for X .

The second ODE

$$\frac{d^2X}{dx^2} = CX = -\lambda^2 X$$

inherits the boundary conditions from the original PDE: $X(0) = 0 = X(L)$. This is a constant coefficient equation so we look for the auxiliary polynomial p . To avoid overloading the symbol λ let us take $p = p(r)$. In this case

$$p(r) = r^2 + -C.$$

The roots of the auxiliary polynomial are $r = \pm\sqrt{C}$. There are three options here.

1. C is negative, i.e. $C = -\lambda^2$. Then the solutions are

$$X(x) = A \cos(\lambda x) + B \sin(\lambda x).$$

The boundary condition $X(0) = 0$ forces the constant $A = 0$. The second boundary condition implies that

$$B \sin(\lambda L) = 0$$

which has a non-trivial solution only if $\lambda L = n\pi$ for integer n . This can be rearranged to show that $\lambda = \frac{n\pi}{L}$ and the corresponding solutions are

$$X(x) = B \sin\left(\frac{n\pi x}{L}\right).$$

It's often taken just as $-\lambda$ but that requires working with a square root later on. Be aware of this slightly non-standard choice we've made if you are reading complementary material.

2. $C = 0$. We have a repeated root and the solutions are $X(x) = A + Bx$. The boundary conditions force $A = B = 0$. This is the trivial solution, so we discard it.
3. C is positive, i.e. $C = \lambda^2$. This leads to two distinct real roots for the auxiliary polynomial and the solutions are therefore

$$X(x) = Ae^{\lambda x} + Be^{-\lambda x}.$$

The first boundary condition is satisfied by $A + B = 0$. The second leads to $Ae^{2\lambda L} + B = 0$. Subtracting the two from each other gives $A(e^{2\lambda L} - 1) = 0$ which forces $A = 0$. This in turn forces $B = 0$ and the only solution is the trivial one.

We have recovered the condition that C must be negative, i.e. $C = -\lambda^2$ for nontrivial solutions to the heat equation to be possible. Before we put the time- and space-dependent solutions together to recover the solution to the heat equation we make the following observation. If we rewrite the boundary-value problem as

$$D^2(X) = -\lambda^2 X$$

we recognise it as an eigenvalue problem which has eigenvalues $-\lambda^2 = \frac{n\pi}{L}$ and corresponding eigenfunctions $\sin\left(\frac{n\pi x}{L}\right)$.

The solutions to the heat equation were assumed to have the form $u(x, t) = X(x)T(t)$. We solved above for the functions X and T . Putting it together we see that

$$u(x, t) = C \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t}.$$

If the initial condition for our differential equation is in exactly the right form, i.e. it matches the expression above for suitably chosen C and n when $t = 0$, then we're in luck. We can solve the differential equation. If, for example, $u_0 = 5 \sin\left(\frac{3\pi x}{L}\right)$ then

$$u(x, t) = 5 \sin\left(\frac{3\pi x}{L}\right) e^{-k(3\pi/L)^2 t}$$

and we can kick up our feet and enjoy a beer. Most of the time we are not so fortunate. We discuss what to do when we're unlucky in the next section on Fourier series. Before we proceed there is one final remark. In the case of homogeneous boundary conditions of the second type we can do the same and we will discover that the solution is either

$$u(x, t) = C \cos\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t}$$

corresponding to a nonzero separation constant or

$$u(x, t) = C$$

corresponding to a zero separation constant.

Linearity, Superposition and Orthogonality – the Foundations of Fourier Series

The heat equation is linear¹. This means that a linear combination of two solutions to the heat equation is still a solution to the heat equation, and the boundary and initial conditions satisfied by this linear combination is just the same linear combination of the boundary and initial conditions satisfied by those two equations.

¹ Another tutorial problem.

In the case of the homogeneous Dirichlet boundary conditions described above this leads to the solution

$$u(x, t) = \sum_{n=1}^N C_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t}$$

where each contribution to the solution could have a different amplitude C_n . This allows us to solve the heat equation for an initial condition of the form

$$u(x, 0) = \sum_{n=1}^N C_n \sin\left(\frac{n\pi x}{L}\right)$$

i.e. if the initial condition for our problem can be written as a finite sum of appropriate sine functions. In fact, something much more useful (and perhaps harder to believe) is true. The infinite sum

$$u(x, t) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t}$$

is a solution to the heat equation with homogeneous Dirichlet boundary conditions, and any “reasonable” initial condition can be expressed as an infinite sum of the form

$$u(x, 0) = u_0(x) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right).$$

These claims, and the “reasonableness” of initial conditions, will occupy us for the next while. We are not going to attempt a rigorous proof for the convergence of these series, but nor are we simply going to provide a recipe for calculating the unknown coefficients C_n from a specified initial condition. We will prove as much as we can with the tools at our disposal, but if you really want to understand Theorem 4.1 you’ll have to take at least another year of mathematics.

Vector spaces may be equipped with an inner product (e.g. the dot product in $\mathbb{R}^n$). We introduce an inner product for the vector space of real-valued functions defined on some interval. The **inner product** of two real-valued functions f and g defined on an interval $[0, L]$ is

$$\langle f, g \rangle = \int_0^L f(x)g(x) \, dx.$$

In much the same way that two vectors are orthogonal to each other in $\mathbb{R}^n$ if their dot product is zero, we call two functions **orthogonal** if their inner product is zero, i.e. if $\langle f, g \rangle = 0$.

EXAMPLE. Show that the functions $f_m(x) = \sin\left(\frac{m\pi x}{L}\right)$ and $f_n(x) = \sin\left(\frac{n\pi x}{L}\right)$ are orthogonal if $m \neq n$. Determine also the inner product $\langle f_n, f_n \rangle$.

$$\begin{aligned}\langle f_n, f_m \rangle &= \int_0^L \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx \\ &= \frac{1}{2} \int_0^L \cos\left(\frac{(m-n)\pi x}{L}\right) - \cos\left(\frac{(m+n)\pi x}{L}\right) dx \\ &= \frac{1}{2} \left[\frac{L}{(m-n)\pi} \sin\left(\frac{(m-n)\pi x}{L}\right) - \frac{L}{(m+n)\pi} \sin\left(\frac{(m+n)\pi x}{L}\right) \right]_0^L \\ &= \frac{1}{2} \left(\frac{L}{(m-n)\pi} \sin((m-n)\pi) - \frac{L}{(m+n)\pi} \sin((m+n)\pi) \right) \\ &= 0\end{aligned}$$

So the functions f_m and f_n are orthogonal. If $m = n$ then

$$\begin{aligned}\langle f_n, f_n \rangle &= \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx \\ &= \frac{1}{2} \int_0^L 1 - \cos\left(\frac{2n\pi x}{L}\right) dx \\ &= \frac{L}{2} - \left[\frac{L}{2n\pi} \sin\left(\frac{2n\pi x}{L}\right) \right]_0^L \\ &= \frac{L}{2}\end{aligned}$$

These results can be more compactly rewritten as

$$\langle f_m, f_n \rangle = \begin{cases} 0 & \text{if } m \neq n \\ \frac{L}{2} & \text{if } m = n \end{cases}.$$

At this point it is useful to make the observation that although we haven't gotten any closer to explaining why our "reasonable" initial condition can be expressed as the infinite sum

$$u_0(x) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right)$$

or what makes an initial condition reasonable, we can use the orthog-

onality of sines to determine the unknown constants C_n .

$$\begin{aligned}
 u_0(x) &= \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right) \\
 u_0(x) \sin\left(\frac{m\pi x}{L}\right) &= \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) \\
 \int_0^L u_0(x) \sin\left(\frac{m\pi x}{L}\right) dx &= \int_0^L \left(\sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) \right) dx \\
 \langle u_0, f_m \rangle &= \sum_{n=1}^{\infty} \left(C_n \int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx \right) \\
 \langle u_0, f_m \rangle &= \sum_{n=1}^{\infty} C_n \langle f_m, f_n \rangle \\
 \langle u_0, f_m \rangle &= C_m \frac{L}{2} \\
 \Rightarrow C_m &= \frac{2}{L} \langle u_0, f_m \rangle
 \end{aligned}$$

There are some important details that we are glossing over for now² but let's take a moment to actually solve a problem using this result before we get back to the gritty details.

² Is this term-by-term integration valid?
Can we simply exchange the integral and the summation?

EXAMPLE. Solve the heat equation

$$u_t = ku_{xx}$$

with homogeneous Dirichlet boundary conditions and the initial condition

$$u(x, 0) = x \text{ for } 0 < x < L.$$

From the previous work we know that our solution has the form

$$u(x, t) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t}.$$

The only thing we don't know about this solution is the value of the constants C_n which we determine using

$$C_m = \frac{2}{L} \langle x, f_m \rangle.$$

In this case we have that

$$\begin{aligned}
 C_m &= \frac{2}{L} \int_0^L x \sin\left(\frac{m\pi x}{L}\right) dx \\
 &= \frac{2}{L} \left[-x \frac{L}{m\pi} \cos\left(\frac{m\pi x}{L}\right) + \frac{L}{m\pi} \int \cos\left(\frac{m\pi x}{L}\right) dx \right]_0^L \\
 &= \frac{2}{L} \left[-x \frac{L}{m\pi} \cos\left(\frac{m\pi x}{L}\right) + \left(\frac{L}{m\pi}\right)^2 \sin\left(\frac{m\pi x}{L}\right) \right]_0^L \\
 &= (-1)^{1+m} \frac{2L}{m\pi}.
 \end{aligned}$$

Our solution is therefore

$$u(x, t) = \sum_{n=1}^{\infty} (-1)^{1+n} \frac{2L}{n\pi} \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t}.$$

Fourier Series

A function f is **piecewise smooth** (on an interval) if that interval can be broken up into a finite number of subintervals and on the interior of each of those subintervals the function f and its derivative f' are continuous.

The **periodic extension** of a function f defined on an interval $(-L, L]$ to $\mathbb{R}$ is defined by imposing the condition

$$f(x) = f(x + 2Ln)$$

where n is the integer chosen so that $x + 2Ln \in (-L, L]$. The function f is repeated and shifted by $2L$ along the real axis. The pictures below may make this clearer.

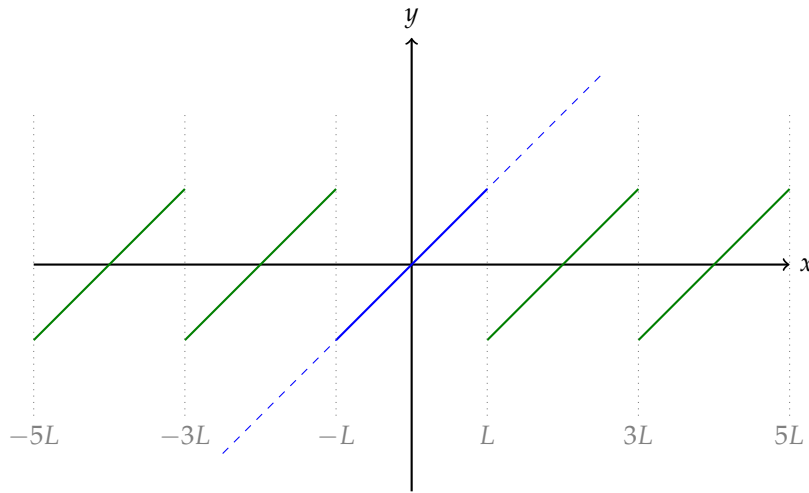


Figure 4.2: The periodic extension of the function $f(x) = x$ defined on $(-L, L]$. The function f is plotted in blue. The dashed blue curve is the behaviour of the function f for $x > L$ and the periodic extension of f is plotted in green.

The **Fourier series** of a function f defined on the interval $[-L, L]$ is the infinite series

$$f(x) \sim A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

where the **Fourier coefficients** are defined by

$$\begin{aligned} A_0 &= \frac{1}{2L} \langle f(x), 1 \rangle, \\ A_n &= \frac{1}{L} \langle f(x), \cos\left(\frac{n\pi x}{L}\right) \rangle, \\ B_n &= \frac{1}{L} \langle f(x), \sin\left(\frac{n\pi x}{L}\right) \rangle. \end{aligned}$$

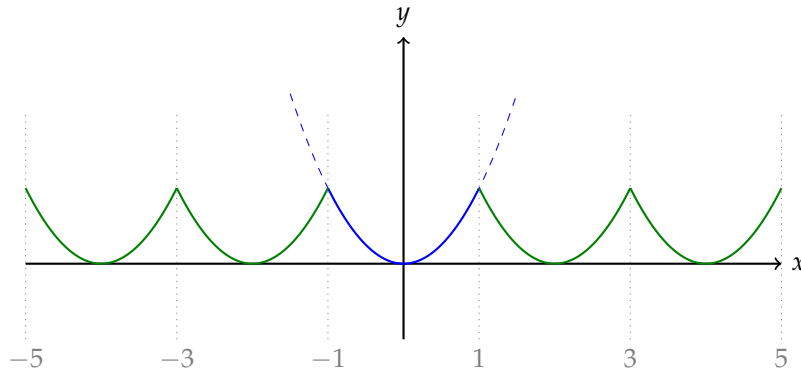


Figure 4.3: The periodic extension of the function $f(x) = x^2$ defined on $(-1, 1]$. The function f is plotted in blue. The dashed blue curve is the behaviour of the function f for $x > |L|$ and the periodic extension of f is plotted in green.

Note that the inner product in this definition is defined as the integral over the interval $[-L, L]$, not just $[0, L]$ as we did earlier. We have used the symbol $\sim$ to indicate that the right-hand side is the Fourier series of the left-hand side, but there is (in general) no guarantee that the Fourier series is convergent, and if it is convergent, that it converges to f .

THEOREM 4.1 (Fourier's Theorem). *If the function f is piecewise smooth on the interval $[-L, L]$ then the Fourier series of f converges to*

1. *the periodic extension of $f(x)$ where the periodic extension is continuous,*
2. *the average of the two one-sided limits at points where the periodic extension has a jump discontinuity.*

If the function f is continuous on the interval $[-L, L]$ then for $x \in (-L, L)$ we can write

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right).$$

If f is not just continuous, but $f(-L) = f(L)$ as well, then the periodic extension of f has no discontinuities and f is equal to its Fourier series on the closed interval $[-L, L]$.

There is an important conceptual point that has not been made explicit in either this or the preceding section. Fourier's theorem, and the previous results about orthogonality, tell us that the vector space of continuous functions with a period of $2L$ has a basis

$$\left\{1, \cos\left(\frac{\pi x}{L}\right), \sin\left(\frac{\pi x}{L}\right), \cos\left(\frac{2\pi x}{L}\right), \sin\left(\frac{2\pi x}{L}\right), \dots\right\}.$$

These functions constitute an infinite-dimensional vector space.

Just as any function f can be rewritten as the sum of an even and odd function, any Fourier series

$$A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

can be split into two constituent pieces. An odd Fourier sine series

$$\sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

and an even Fourier cosine series.

$$A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right).$$

The Fourier sine and cosine series arise naturally when we solve the heat equation, so this decomposition is not an artificial one. In fact, any piecewise smooth function defined on $[0, L]$ can be expressed as either a Fourier sine *or* a Fourier cosine series. That sounds a little strange, since the Fourier sine series is odd and the Fourier cosine series is even, so it's not clear how the same function (in our case, the same initial condition to the heat equation) can be expressed in these two seemingly contradictory ways. We illustrate this idea with an example.

EXAMPLE. Find the Fourier sine series for the initial condition $u(x, 0) = x$ on $[0, L]$. Find the Fourier cosine series for the same initial condition. Check that the conditions for convergence described above are satisfied and explain what the Fourier series are converging to at each point in $[0, L]$.

For the Fourier sine series we start by creating the **odd extension** of $u(x, 0)$. Let

$$f(x) = \begin{cases} u(x, 0) & \text{for } 0 \leq x \leq L \\ -u(-x, 0) & \text{for } -L \leq x < 0 \end{cases}$$

which for the simple initial condition we are discussing reduces to $f(x) = x$. Now we consider its periodic extension (plotted previously in Figure 4.2). Since f is piecewise smooth on the interval $[-L, L]$ its Fourier series is convergent by Theorem 4.1 so

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

on the open intervals $(-L \pm 2kL, L \pm 2kL)$ for $k \in \mathbb{N}$ and at $x = kL$ it converges to 0 (the average of the limits from the left and from the right at that point). The coefficients of this series are

$$B_n = (-1)^{1+n} \frac{2L}{n\pi}$$

Actually, we did a slightly different calculation. We previously defined the inner product over $[0, L]$, here it's over $[-L, L]$. You could check that the result is the same no matter which way you do it.

which we calculated earlier, and

$$\begin{aligned} A_n &= \frac{1}{L} \langle f(x), \cos\left(\frac{n\pi x}{L}\right) \rangle \\ &= \frac{1}{L} \int_{-L}^L x \cos\left(\frac{n\pi x}{L}\right) dx \\ &= 0 \end{aligned}$$

since we are integrating an odd function. Similarly $A_0 = 0$. Putting it all together we have that

$$u(x, 0) = \sum_{n=1}^{\infty} (-1)^{1+n} \frac{2L}{n\pi} \sin\left(\frac{n\pi x}{L}\right)$$

on $[0, L)$ and at $x = L$ the series converges to 0. The partial sums of this series are plotted in Figure 4.4.

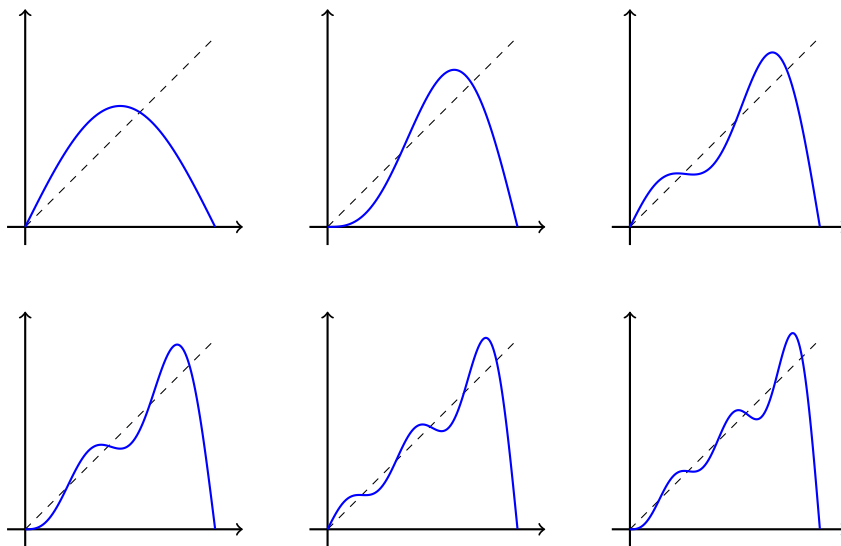


Figure 4.4: Partial sums of the Fourier sine series for x on the interval $[0, L]$. The top left figure shows the partial sum with one term. Successive figures (left to right, top to bottom) show the partial sum with two, three, four, five and six terms.

For the Fourier cosine series we start by creating the **even extension** of $u(x, 0)$. Let

$$g(x) = \begin{cases} u(x, 0) & \text{for } 0 \leq x \leq L \\ u(-x, 0) & \text{for } -L \leq x < 0 \end{cases}$$

which, in this very simple case, can be simplified to $g(x) = |x|$. Again, we consider the periodic extension of g (which is a triangular wave). This time g is continuous and its periodic extension has no jump discontinuities so

$$g(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

everywhere. The coefficients B_n are all zero since $B_n = \frac{1}{L} \langle g(x), \sin\left(\frac{n\pi x}{L}\right) \rangle$ which is an integral of an odd function with symmetric limits of integration. The remaining coefficients are

$$\begin{aligned} A_0 &= \frac{1}{2L} \langle |x|, 1 \rangle \\ &= \frac{1}{2L} \int_{-L}^L |x| \, dx \\ &= \frac{L}{2} \end{aligned}$$

and

$$\begin{aligned} A_n &= \frac{1}{L} \langle |x|, \cos\left(\frac{n\pi x}{L}\right) \rangle \\ &= \frac{1}{L} \int_{-L}^L |x| \cos\left(\frac{n\pi x}{L}\right) \, dx \\ &= \frac{2}{L} \int_0^L x \cos\left(\frac{n\pi x}{L}\right) \, dx \\ &= \frac{2}{L} \left[x \left(\frac{L}{n\pi} \right) \sin\left(\frac{n\pi x}{L}\right) - \left(\frac{L}{n\pi} \right) \int \sin\left(\frac{n\pi x}{L}\right) \, dx \right]_0^L \\ &= \frac{2}{L} \left[\left(\frac{L}{n\pi} \right)^2 \cos\left(\frac{n\pi x}{L}\right) \right]_0^L \\ &= \frac{2L}{(n\pi)^2} ((-1)^n - 1). \end{aligned}$$

The Fourier cosine series is therefore

$$u(x, 0) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2L}{(n\pi)^2} ((-1)^n - 1) \cos\left(\frac{n\pi x}{L}\right)$$

and this is convergent for all $x \in [0, L]$. The partial sums of this series are plotted in Figure 4.5.

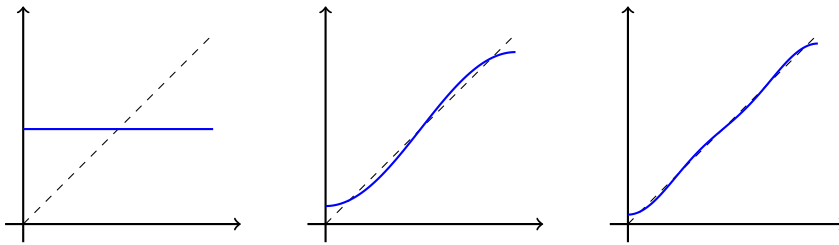


Figure 4.5: Partial sums of the Fourier cosine series for x on the interval $[0, L]$. The leftmost figure shows the partial sum with one term. Successive figures show the partial sum with two and three terms.

Term-by-term Integration and Differentiation

In our discussion of Fourier series so far we have glossed over three important points.

1. The missing proof for Theorem 4.1. We're going to keep glossing over this. If you want a better answer consider majoring in mathematics.
2. An earlier claim that the infinite sum

$$u(x, t) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t}$$

is a solution to the heat equation with homogeneous Dirichlet boundary conditions.

3. The calculation of the Fourier coefficients by taking an inner product between our initial condition and an infinite sum, i.e. integrating an infinite sum.

We are not in a position to address the first point, but we can say something useful about the second and third point. If we were to substitute in the expression for u from point 2. into the heat equation we would have

$$\begin{aligned} \frac{\partial}{\partial t} \left(\sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t} \right) &= k \frac{\partial^2}{\partial x^2} \left(\sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t} \right) \\ \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right) \frac{\partial}{\partial t} \left(e^{-k(n\pi/L)^2 t} \right) &= k \sum_{n=1}^{\infty} C_n \frac{\partial^2}{\partial x^2} \left(\sin\left(\frac{n\pi x}{L}\right) \right) e^{-k(n\pi/L)^2 t} \\ -k \left(\frac{n\pi}{L} \right)^2 \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t} &= -k \left(\frac{n\pi}{L} \right)^2 \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t} \end{aligned}$$

which is true. This seems like we've shown that it is a solution, but in doing so we've exchanged an infinite sum and a derivative in the second step of our calculation. That is, in general, a bad thing. For arbitrary functions f_i there is no guarantee that

$$\frac{d}{dx} \left(\sum_{i=1}^{\infty} f_i(x) \right) = \sum_{i=1}^{\infty} \frac{d}{dx} (f_i(x)).$$

Even if the original sum converges the sum of the derivatives might not, and even if the sum of the derivatives converges, it's not necessarily to the derivative of the sum. Theorem 4.2 below explains when it's safe to do this term-by-term differentiation of Fourier series.

THEOREM 4.2 (Differentiating Fourier Sine Series). *If f is an odd, continuous function defined on $[-L, L]$, f' is piecewise smooth and the Fourier sine series of f is continuous (i.e. if $f(-L) = f(L)$) then the Fourier sine series can be differentiated term-by-term and the resulting series is the Fourier cosine series of the derivative f' (but it might not converge to f').*

Let f be an odd, continuous function defined on $[-L, L]$ with piecewise smooth derivative f' . Then

$$f \sim \sum_{i=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right).$$

Since the derivative of an odd function is even we must have

$$f' \sim A_0 + \sum_{i=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right).$$

If we were to differentiate the Fourier series of f term-by-term we would have

$$f' \sim \sum_{i=1}^{\infty} B_n \left(\frac{n\pi}{L}\right) \cos\left(\frac{n\pi x}{L}\right).$$

We are going to verify that these two different expressions for the Fourier series of f' are equivalent, i.e. we have to check that $A_0 = 0$ and that $A_n = \frac{n\pi}{L} B_n$.

$$\begin{aligned} A_0 &= \frac{1}{2L} \langle f'(x), 1 \rangle \\ &= \frac{1}{2L} \int_{-L}^L f'(x) \, dx \\ &= \frac{1}{2L} [f]_{-L}^L \\ &= \frac{1}{2L} [f(L) - f(-L)] \end{aligned}$$

which is equal to zero only if $f(L) = f(-L)$. We also see that

$$\begin{aligned} A_n &= \frac{1}{L} \langle f'(x), \cos\left(\frac{n\pi x}{L}\right) \rangle \\ &= \frac{1}{L} \int_{-L}^L f'(x) \cos\left(\frac{n\pi x}{L}\right) \, dx \\ &= \frac{1}{L} \left[f(x) \cos\left(\frac{n\pi x}{L}\right) + \int f(x) \left(\frac{n\pi}{L}\right) \sin\left(\frac{n\pi x}{L}\right) \, dx \right]_{-L}^L \\ &= \frac{1}{L} (-1)^n [f(L) - f(-L)] + \frac{n\pi}{L} \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) \, dx \end{aligned}$$

which is again only equal to $\frac{n\pi}{L} B_n$ if $f(L) = f(-L)$.

EXAMPLE. Consider the function $f(x) = x$ defined on the interval $[-L, L]$. Note that $f(L) \neq f(-L)$ so the conditions above are not satisfied. We will show that term-by-term differentiation in this case does not work. Recall from previous calculations that

$$f(x) \sim \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2L}{n\pi} \sin\left(\frac{n\pi x}{L}\right).$$

Simply differentiating term-by-term gives

$$f'(x) \sim \sum_{n=1}^{\infty} (-1)^{n+1} 2 \cos\left(\frac{n\pi x}{L}\right).$$

This is obviously not the correct cosine series for f' , since f' is simply the constant function $f'(x) = 1$. Its Fourier cosine series is just

$$1 \sim 1.$$

The difference between the real cosine series of f' and the cosine series that we found by differentiating term-by-term is exactly equal to the terms

$$\frac{1}{2L} [f(L) - f(-L)] \text{ and } \frac{1}{L} (-1)^n [f(L) - f(-L)]$$

that appeared in the proof above.

COROLLARY 4.3. *If f is an odd, continuous function with Fourier sine series*

$$f \sim \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

and f' is piecewise smooth, then f' has the cosine series

$$f' \sim \frac{1}{2L} [f(L) - f(-L)] + \sum_n \left(B_n \frac{n\pi}{L} + \frac{1}{L} (-1)^n [f(L) - f(-L)] \right) \cos\left(\frac{n\pi x}{L}\right).$$

Although we will not prove it, a similar result holds for term-by-term differentiation of the Fourier cosine series of an even function. Here life is a little simpler since the cosine series will always satisfy the condition that $f(L) = f(-L)$. One final observation is that any function can be decomposed into even and odd components and so the restriction in Theorem 4.2 that we consider only odd functions is not very restrictive at all. One final example illustrates the problem with convergence, even when term-by-term differentiation works.

EXAMPLE. Consider the function $g(x) = x^2$ defined on $[-L, L]$. It converges to its Fourier series,

$$g(x) = \frac{L^2}{3} + \sum_{n=1}^{\infty} \frac{4L^2}{n^2\pi^2} (-1)^n \cos\left(\frac{n\pi x}{L}\right),$$

but even in this very nice case there is no convergence of the Fourier series of its derivative. That Fourier series can be calculated either by term-by-term differentiation or directly. We have

$$g'(x) \sim \sum_{n=1}^{\infty} \frac{4L}{n\pi} (-1)^{n+1} \sin\left(\frac{n\pi x}{L}\right)$$

where the problem with convergence is at the endpoints. Although $g'(L)$ and $g'(-L)$ could be defined using left- and right-derivatives

at those points, the periodic extension of g is not differentiable there, and the Fourier series of g' converges to zero at those points.

Now we turn to term-by-term integration of Fourier series. This is important because it's how we actually calculate the coefficients of the series. Recall that we used the orthogonality of sines and cosines with respect to the inner product to eliminate all but one of the terms in the series. That allowed us to determine the value of the remaining coefficient. This inner product is really just integration so we should check if and when term-by-term integration is valid.

THEOREM 4.4 (Integrating Fourier Series). *If f is a piecewise smooth function defined on $[-L, L]$ its Fourier series can be integrated term-by-term and the resulting infinite series converges to the integral of f on $[-L, L]$ (but it may not be the Fourier series of the integral of f).*

Consider a piecewise smooth function f with Fourier series

$$f(x) \sim a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right).$$

If we were to simply integrate term-by-term we would find that

$$\begin{aligned} \int_{-L}^x f(s) \, ds &\sim a_0 \int_{-L}^x 1 \, ds + \sum_{n=1}^{\infty} \left(a_n \int_{-L}^x \cos\left(\frac{n\pi s}{L}\right) \, ds \right) + \sum_{n=1}^{\infty} \left(b_n \int_{-L}^x \sin\left(\frac{n\pi s}{L}\right) \, ds \right) \\ \int_{-L}^x f(s) \, ds &\sim a_0(x+L) + \sum_{n=1}^{\infty} a_n \frac{L}{n\pi} \sin\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \frac{L}{n\pi} \left((-1)^n - \cos\left(\frac{2\pi x}{L}\right) \right) \end{aligned}$$

We show below that this expression converges at every point in $[-L, L]$, i.e. we could write $=$ rather than $\sim$. First define the function

$$F(x) = \int_{-L}^x f(s) \, ds.$$

Since f is piecewise smooth the function F is continuous. Note, however, that there is no guarantee that F has a continuous Fourier series since

$$F(-L) = 0$$

and

$$F(L) = \int_{-L}^L f(t) \, dt = 2a_0L$$

By the definition of a_0 .

which is in general not equal to zero. We introduce a new function G which is obtained from F simply by subtracting the straight line that passes through the points $(-L, 0)$ and $(L, 2a_0L)$, i.e.

$$G(x) = F(x) - a_0(x+L).$$

Now $G(-L) = G(L) = 0$, and it's also continuous since it's the difference of two continuous functions, so it has a convergent Fourier

series and we can write

$$G(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right).$$

The function F can therefore be expressed as

$$F(x) = A_0 + a_0(x+L) + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right).$$

Although this is a convergent series it is *not* a Fourier series because of the term a_0x . What we will do now is to find expressions for the coefficients A_0 , A_n and B_n in terms of a_0 , a_n and b_n and show that this matches the expression we determined above by naïvely integrating term-by-term.

From the definition of the Fourier series coefficients we have that

$$\begin{aligned} A_n &= \frac{1}{L} \langle G, \cos\left(\frac{n\pi x}{L}\right) \rangle \\ &= \frac{1}{L} \int_{-L}^L G(x) \cos\left(\frac{n\pi x}{L}\right) dx \\ &= \frac{1}{L} \int_{-L}^L (F(x) - a_0(x+L)) \cos\left(\frac{n\pi x}{L}\right) dx \\ &= \frac{1}{L} \int_{-L}^L (F(x) - a_0L) \cos\left(\frac{n\pi x}{L}\right) dx && \text{Integrate by parts.} \\ &= -\frac{1}{L} \frac{L}{n\pi} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \\ &= -\frac{L}{n\pi} b_n. \end{aligned}$$

A similar calculation shows that

$$B_n = \frac{L}{n\pi} a_n.$$

Finally, we can calculate the value of A_0 by simply evaluating G at $x = -L$. Since $G(-L) = 0$ we recover

$$\begin{aligned} 0 &= A_0 + \sum_{n=1}^{\infty} A_n \cos(n\pi) \\ \Rightarrow A_0 &= -\sum_{n=1}^{\infty} A_n (-1)^n \\ &= \sum_{n=1}^{\infty} (-1)^n \frac{L}{n\pi} b_n. \end{aligned}$$

Substitution into the expression for F above gives

$$\int_{-L}^x F(s) ds = F(x) = a_0(x+L) + \sum_{n=1}^{\infty} a_n \frac{L}{n\pi} \sin\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \frac{L}{n\pi} \left((-1)^n - \cos\left(\frac{2\pi x}{L}\right) \right).$$

Term-by-term integration of a Fourier series produces another convergent series.

The Fourier Transform

The work we've done above has been for the solution to the heat equation on a finite domain, i.e. for $x \in [0, L]$. Now we turn our attention to the heat equation on an infinite domain, i.e.

$$\begin{aligned} \frac{\partial u}{\partial t} &= k \frac{\partial^2 u}{\partial x^2} \quad \text{for } -\infty < x < \infty \text{ and } 0 < t & \text{PDE} \\ u(x, 0) &= u_0(x) & \text{IC} \end{aligned}$$

Although it appears that there are no boundary conditions, physical considerations suggest that we should have

$$\lim_{x \rightarrow -\infty} u(x, t) = 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} u(x, t) = 0$$

for all t . This is a necessary but not sufficient condition to ensure that the totally energy of the system is finite.

In the finite case, we solved the heat equation by separating variables. The resulting time-dependent ODE has simple exponential solutions, but the space-dependent ODE was a little trickier. We saw that the solutions were combinations of sines and cosines, but these trigonometric functions had strictly controlled frequencies. Really we had solved an eigenvalue problem and found a discrete set of eigenvalues

$$-\lambda^2 = \frac{n\pi}{L}$$

with corresponding eigenfunctions

$$\sin\left(\frac{n\pi x}{L}\right) \quad \text{and} \quad \cos\left(\frac{n\pi x}{L}\right).$$

Finally we spent some time discussing how our initial condition might be expressed as an infinite Fourier series

$$u_0(x) \sim a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right).$$

As a first step toward solving the heat equation on an infinite domain we are going to write our Fourier series more compactly using complex exponentials. It can be shown³ that the Fourier series above can be expressed as

³ It's a tutorial problem.

$$u_0(x) \sim \sum_{n=-\infty}^{\infty} c_n e^{-i\frac{n\pi}{L}x}$$

where the coefficients c_n are given by

$$c_n = \frac{1}{2L} \int_{-L}^L u_0(x) e^{i\frac{n\pi}{L}x} dx.$$

The idea is that we will take the limit as $L \rightarrow \infty$. The eigenvalues

$$-\lambda^2 = \frac{n\pi}{L}$$

become more tightly packed together as L increases and in the limit we recover a continuous spectrum of eigenvalues rather than a discrete set. The Fourier series, which is a sum over these discrete eigenvalues, turns into the Fourier transform, which is an integration over the continuous spectrum of eigenvalues.

Introducing the Fourier Transform

Let $\omega = -\lambda^2 = \frac{n\pi}{L}$ and f be a piecewise smooth, continuous function on $[-L, L]$ such that $f(-L) = f(L) = 0$. Then

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{-i\omega x}$$

where

$$c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{i\omega x} dx.$$

Substituting the expression for c_n back into the expression for f gives

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{1}{2L} \left(\int_{-L}^L f(s) e^{i\omega s} ds \right) e^{-i\omega x}$$

where we have changed the variable of integration in the definition of c_n from x to s to prevent overloading the symbol x . Since

$$\omega = \frac{n\pi}{L}$$

we have that

$$\Delta\omega = \frac{(n+1)\pi}{L} - \frac{n\pi}{L} = \frac{\pi}{L}$$

so the expression for f becomes

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{1}{2\pi} \Delta\omega \left(\int_{-L}^L f(s) e^{i\omega s} ds \right) e^{-i\omega x}.$$

As we take the limit $L \rightarrow \infty$ two things happen. First, the integral from $-L$ to L is replaced by the integral from $-\infty$ to ∞ . Second, the value $\Delta\omega \rightarrow 0$ so the sum

$$\sum_{n=-\infty}^{\infty} \frac{1}{2\pi} \Delta\omega \left(\int_{-\infty}^{\infty} f(s) e^{i\omega s} ds \right) e^{-i\omega x}$$

can be understood as a Riemann sum, i.e. an integral, and so we can write

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(s) e^{i\omega s} ds \right) e^{-i\omega x} d\omega.$$

This is the **Fourier integral identity**. Our derivation has not been completely rigorous. When we evaluate a limit everything changes at the same time. We shouldn't first change the finite integral to

an infinite integral and only after that change the infinite sum to an integral. For a well-behaved function like f it doesn't make any difference but it's worth noting that this is not a formal proof of the Fourier integral identity.

The Fourier integral identity is more often written as a pair of equations relating a function f to its **Fourier transform** $\hat{f}$.

$$\hat{f}(\omega) = \mathcal{F}[f] = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{i\omega x} dx \quad (4.1)$$

$$f(x) = \mathcal{F}^{-1}[\hat{f}] = \int_{-\infty}^{\infty} \hat{f}(\omega) e^{-i\omega x} d\omega \quad (4.2)$$

Equation 4.1 above defines the Fourier transform of a function f . It converts a function from the spatial domain to the frequency domain. Equation 4.2 defines the inverse Fourier transform. It reconstructs a function in the spatial domain from information in the frequency domain. Note that this is not the only way the Fourier transform may be defined, so be careful when reading other sources.

The motivation for introducing the Fourier transform is illustrated in Figure 4.6. Given a partial differential equation for a function $u(x, t)$ that depends on space and time, e.g. the heat equation, we can convert it into an ordinary differential equation by applying the Fourier transform. This ordinary differential equation describes a function $\hat{u}(\omega, t)$ but the dependence on ω in the equation is purely algebraic. The only derivatives are derivatives with respect to t . We solve this equation and then take the inverse transform of $\hat{u}(\omega, t)$ to recover a solution to the original partial differential equation. We discuss the details in the next section.

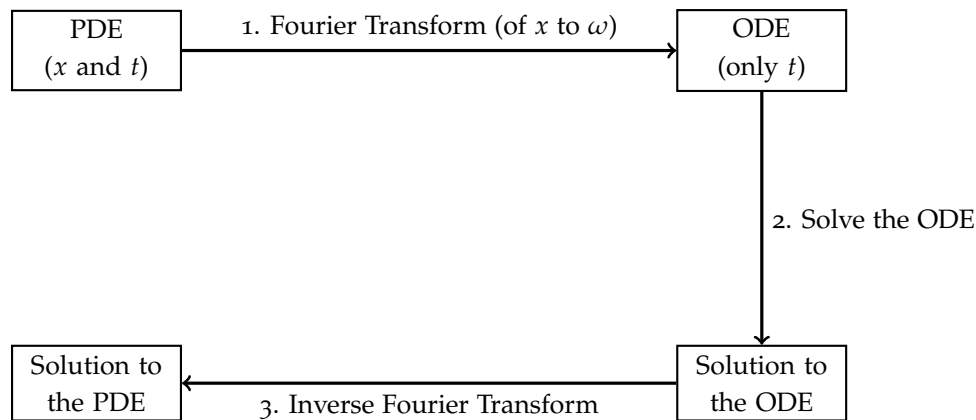


Figure 4.6: Using the Fourier transform to solve a partial differential equation.

Solving the heat equation with the Fourier transform

There are three things we must understand before we are going to be able to implement the scheme outlined by Figure 4.6.

1. If a function $f(x, t)$ is transformed to $\hat{f}(\omega, t)$, how are the partial derivatives $f_x(x, t)$, $f_{xx}(x, t)$ and so on transformed?
2. What is the Fourier transform of a Gaussian, i.e. if

$$f(x) = e^{-ax^2}$$

what is $\hat{f}(\omega)$?

3. If $\hat{f}(\omega)$ and $\hat{g}(\omega)$ are the Fourier transforms of $f(x)$ and $g(x)$, can we recover a function $h(x)$ if we know that $\hat{h}(\omega) = \hat{f}(\omega)\hat{g}(\omega)$?

We answer these questions below with two lemmas and a theorem.

LEMMA 4.5. *The Fourier transform (from x to ω) of the partial derivatives with respect to x is⁴ given by*

$$\mathcal{F} \left[\frac{\partial^n f}{\partial x^n} \right] = (-i\omega)^n \hat{f}(\omega, t).$$

⁴ Provided the derivatives all decay sufficiently quickly and are continuous.

Proof. The transform of a derivative can be calculated easily by using the definition of the transform and integrating by parts.

$$\begin{aligned} \mathcal{F}[f_x] &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f_x(x, t) e^{i\omega x} dx \\ &= \frac{1}{2\pi} \left[f(x, t) e^{i\omega x} \right]_{-\infty}^{\infty} - \frac{i\omega}{2\pi} \int_{-\infty}^{\infty} f(x, t) e^{i\omega x} dx \\ &= -i\omega \mathcal{F}[f] \\ &= -i\omega \hat{f}(\omega, t) \end{aligned}$$

Here we have used the fact that f must decay as $x \rightarrow \pm\infty$ since the Fourier transform is otherwise not well-defined. Calculating the transforms of higher-order derivatives is easy:

$$\mathcal{F}[f_{xx}] = (-i\omega) \mathcal{F}[f_x] = (-i\omega)^2 \hat{f}(\omega, t) = -\omega^2 \hat{f}(\omega, t).$$

This argument can be extended by mathematical induction to the n -th derivative. □

With this in mind, consider what happens when we transform the heat equation.

$$\begin{aligned} u_t &= ku_{xx} \\ \mathcal{F}[u_t] &= \mathcal{F}[ku_{xx}] \\ \hat{u}_t(\omega, t) &= -k\omega^2 \hat{u}(\omega, t) \\ \Rightarrow \hat{u}(\omega, t) &= \hat{u}_0(\omega) e^{-k\omega^2 t} \end{aligned}$$

Here the heat equation, a PDE, is converted into a linear ODE in t which we can solve. The arbitrary constant that appears when we

solve this ODE is an arbitrary constant with respect to the variable t , i.e. it is a function of ω . In fact, it's clear that $\hat{u}_0(\omega)$ is just the Fourier transform of the initial condition $u_0(x)$ of the original heat equation. We would like to recover the solution $u(x, t)$ from it's Fourier transform, i.e.

$$\begin{aligned} u(x, t) &= \int_{-\infty}^{\infty} \hat{u}(\omega, t) e^{-i\omega x} d\omega \\ &= \int_{-\infty}^{\infty} \hat{u}_0(\omega) e^{-k\omega^2 t} e^{-i\omega x} d\omega. \end{aligned} \quad (4.3)$$

This is not an obvious thing to do. To proceed we will first have to answer questions 2. and 3. that we posed above.

LEMMA 4.6. *The (inverse) Fourier transform of a Gaussian is another Gaussian. More precisely:*

$f(x)$	$\hat{f}(\omega)$
$e^{-\beta x^2}$	$\frac{1}{\sqrt{4\pi\beta}} e^{-\omega^2/4\beta}$
$\sqrt{\frac{\pi}{\alpha}} e^{-x^2/4\alpha}$	$e^{-\alpha\omega^2}$

Proof. Let f be the function that has a transform $\hat{f}(\omega) = e^{-\alpha\omega^2}$. From the definition of the inverse Fourier transform we have that

$$f(x) = \int_{-\infty}^{\infty} e^{-\alpha\omega^2} e^{-i\omega x} d\omega. \quad (4.4)$$

To show that the function f is also a Gaussian we will show that it satisfies a linear ODE solved by a Gaussian. We start by differentiating the expression above, then simplify the result using integration by parts.

$$\begin{aligned} f'(x) &= \int_{-\infty}^{\infty} e^{-\alpha\omega^2} (-i\omega) e^{-i\omega x} d\omega \\ &= -i \int_{-\infty}^{\infty} \underbrace{\omega e^{-\alpha\omega^2}}_{-1/2\alpha \frac{d}{d\omega} [e^{-\alpha\omega^2}]} e^{-i\omega x} d\omega \\ &= \frac{i}{2\alpha} \int_{-\infty}^{\infty} \frac{d}{d\omega} [e^{-\alpha\omega^2}] e^{-i\omega x} d\omega \\ &= \frac{i}{2\alpha} [e^{-\alpha\omega^2} e^{-i\omega x}]_{-\infty}^{\infty} - \frac{x}{2\alpha} \int_{-\infty}^{\infty} e^{-\alpha\omega^2} e^{-i\omega x} d\omega \\ &= -\frac{x}{2\alpha} f(x) \end{aligned}$$

After rearranging we see that we have a linear first-order differential equation.

$$f'(x) + \frac{x}{2\alpha} f(x) = 0$$

which is easily solved and so

$$f(x) = f_0 e^{-x^2/4\alpha}.$$

The constant f_0 can be determined by setting $x = 0$ in equation 4.4.

$$\begin{aligned} f(0) &= f_0 \\ &= \int_{-\infty}^{\infty} e^{-\alpha\omega^2} d\omega \\ &= \frac{1}{\sqrt{\alpha}} \int_{-\infty}^{\infty} e^{-z^2} dz \\ &= \sqrt{\frac{\pi}{\alpha}} \end{aligned}$$

Putting this together we recover the desired result (at least in one direction) that

$$f(x) = \sqrt{\frac{\pi}{\alpha}} e^{-x^2/4\alpha}.$$

The other direction follows from the linearity of the Fourier transform⁵. Simply let $\beta = \frac{1}{4\alpha}$ in the expressions above and it follows after some simplification. $\square$

⁵ A tutorial problem.

The last question that we have to answer is how to calculate the inverse Fourier transform of a product. To do this we will need the **convolution**. The convolution of two functions f and g is denoted by $f(x) * g(x)$ and is defined as

$$f(x) * g(x) = \int_{-\infty}^{\infty} f(s)g(x-s) ds.$$

It is not difficult to show that the convolution of two functions is commutative, i.e. $f(x) * g(x) = g(x) * f(x)$. With this in mind, we proceed to answer the outstanding question.

THEOREM 4.7 (Convolution Theorem). *Let $\hat{f}(w)$, $\hat{g}(w)$ and $\hat{h}(w)$ be the Fourier transforms of $f(x)$, $g(x)$ and $h(x)$. Then*

$$\hat{h}(w) = \hat{f}(w)\hat{g}(w) \iff h(x) = \frac{1}{2\pi} f(x) * g(x).$$

Proof. If

$$\hat{h}(w) = \hat{f}(w)\hat{g}(w)$$

then

$$\begin{aligned}
h(x) &= \mathcal{F}^{-1} [\hat{h}(w)] \\
&= \mathcal{F}^{-1} [\hat{f}(w)\hat{g}(w)] \\
&= \int_{-\infty}^{\infty} \hat{f}(w)\hat{g}(w)e^{-i\omega x} d\omega \\
&= \int_{-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} f(s)e^{i\omega s} ds \right) \hat{g}(w)e^{-i\omega x} d\omega \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(s)e^{i\omega s} ds \right) \hat{g}(w)e^{-i\omega x} d\omega \\
&\text{now exchange the order of integration} \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(s) \underbrace{\left(\int_{-\infty}^{\infty} \hat{g}(w)e^{-i\omega(x-s)} d\omega \right)}_{g(x-s)} ds \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(s)g(x-s) ds \\
&= \frac{1}{2\pi} f(x) * g(x).
\end{aligned}$$

□

We are finally in a position to put everything together to solve the heat equation. We have already shown that the solution is given by equation 4.3 which we reproduce below

$$u(x, t) = \int_{-\infty}^{\infty} \hat{u}_0(\omega) e^{-k\omega^2 t} e^{-i\omega x} d\omega.$$

The transform of the solution is simply the product of $\hat{u}_0$, the transform of the initial condition, and $e^{-k\omega^2 t}$, a Gaussian. By Lemma 4.6 the inverse transform of this Gaussian is

$$\mathcal{F}^{-1} [e^{-k\omega^2 t}] = \sqrt{\frac{\pi}{kt}} e^{-x^2/4kt}$$

and by Theorem 4.7 the solution is given by

$$\begin{aligned}
u(x, t) &= \frac{1}{2\pi} u_0(x) * \sqrt{\frac{\pi}{kt}} e^{-x^2/4kt} \\
&= u_0(x) * \frac{1}{\sqrt{4\pi kt}} e^{-x^2/4kt} \\
&= \int_{-\infty}^{\infty} u_0(s) \frac{1}{\sqrt{4\pi kt}} e^{-(x-s)^2/4kt} ds \quad (4.5)
\end{aligned}$$

The Gaussian $\frac{1}{\sqrt{4\pi kt}} e^{-(x-s)^2/4kt}$ is often referred to as the **influence function** or **Green's function** for the heat equation. Each linear PDE has its own Green's function, and the solution to the PDE is always a convolution of the Green's function with the initial condition.

Actually evaluating the integral in equation 4.5 can be difficult. Much depends on what exactly the initial condition is, but whatever the initial condition we can make the observation that as $t \rightarrow \infty$ the solution decays. This behaviour is typical of a diffusive equation, and the heat equation is the prototypical diffusion equation. Physically this behaviour makes sense. If a finite amount of energy is imparted to an infinitely large system, eventually the average temperature tends to zero, not matter how hot a small part of the system may have been at the start.

EXAMPLE. Solve the heat equation on the infinite line with the initial condition

$$u_0(x) = \begin{cases} C & |x| \leq L \\ 0 & |x| > L \end{cases}$$

The solution is

$$\begin{aligned} u(x, t) &= \int_{-\infty}^{\infty} u_0(s) \frac{1}{\sqrt{4\pi kt}} e^{-(x-s)^2/4ft} ds \\ &= C \int_{-L}^L \frac{1}{\sqrt{4\pi kt}} e^{-(x-s)^2/4ft} ds. \end{aligned}$$

To evaluate this integral we make the substitution $s = x - \sqrt{4kt}z$ which transforms the integral into

$$\begin{aligned} u(x, t) &= -C \int_{(x+L)/\sqrt{4kt}}^{(x-L)/\sqrt{4kt}} \frac{1}{\sqrt{\pi}} e^{-z^2} dz \\ &= \frac{C}{2} \left(\frac{2}{\sqrt{\pi}} \int_0^{(x+L)/\sqrt{4kt}} e^{-z^2} dz - \frac{2}{\sqrt{\pi}} \int_0^{(x-L)/\sqrt{4kt}} e^{-z^2} dz \right) \\ &= \frac{C}{2} \left(\operatorname{Erf} \left(\frac{x+L}{\sqrt{4kt}} \right) - \operatorname{Erf} \left(\frac{x-L}{\sqrt{4kt}} \right) \right). \end{aligned}$$

This solution has been plotted in Figure 4.7. It's clear that the sharp initial condition is softening and spreading outward. This is typical of a solution to a diffusion equation.

The Black-Scholes Equation

The Black-Scholes equation is a linear PDE that is used to price options. Before we get to the Black-Scholes equation we discuss some of the basics of options.

An **option** is a contract between two parties regarding a possible future transaction of an underlying asset. When the two parties agree to an option, the option buyer pays the option seller. In exchange, the option seller agrees that at a later date they will buy or sell the underlying asset at a predetermined price from/to the option buyer (if such a sale or purchase is profitable for the option buyer).

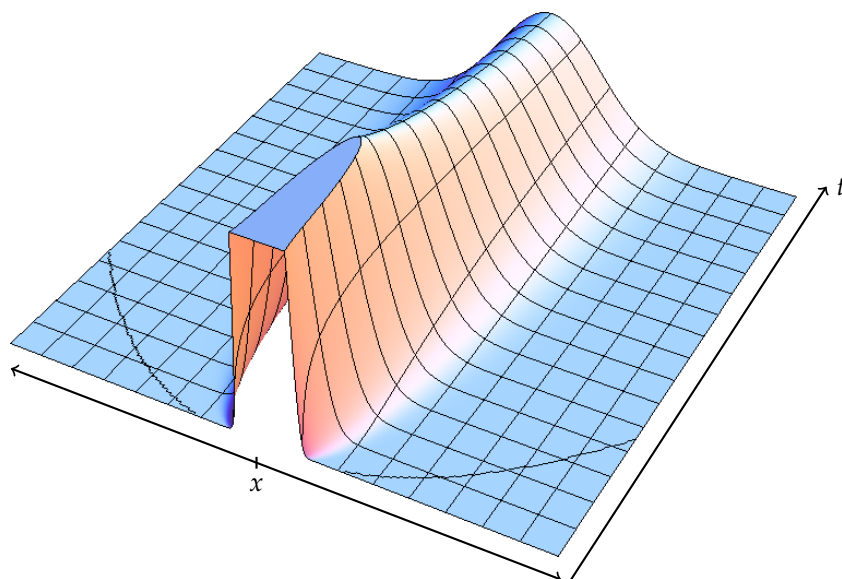


Figure 4.7: The solution to the heat equation with a rectangular pulse initial condition. Here $k = 1$, $L = 1$, $C = 1$. The x axis runs from -10 to 10 and the t axis runs from 0 to 2 .

This predetermined price is called the **strike price**, which may differ from the **spot price** or market value of the asset. The future date of the transaction is the **expiry date** of the option. If the option can be exercised only on the expiry date it is a **European option** and if it can be exercised at any point before the expiry date it is an **American option**. An agreement that the option buyer can sell the asset is a **put option** and an agreement that the option buyer can buy the asset is a **call option**.

Options are used to hedge against the risk associated with a change in the price of the underlying asset, or to speculate on changes in the price of the underlying asset. Some examples will make this clear.

EXAMPLE. Maize currently (in September) trades for R2500 per ton, up from R2000 per ton in January. Willards expects increased demand for Big Corn Bites over the December holiday period, and will need additional maize at the end of November to process into chips. The head of logistics at Willards is worried about the rising cost of maize, so they purchase a European call option for 200 tons of maize from Unigrain with an expiry date of 20 November and a strike price of R2600.

If the spot price of maize is greater than R2600 on 20 November they exercise the call option. Unigrain must sell them 200 tones of maize at R2600, below market value. If the spot price of maize is below R2600 Willards instead lets the option expire and buys 200 tons of maize from the market.

EXAMPLE. Tesla currently trades at \$300 per share. The super-villain Dr Evil buys a put option for 1 million shares of Tesla at \$310 per share, expiring in one week. Two days before expiry he bombs the Tesla factory in California. The Tesla share price drops to \$180 per share. He buys 1 million shares at this low price, and then sells them two days later for \$310 each using the put option.

In both of these examples there is an important piece of information that has been omitted, the option price. How much should Willards pay Unigrain for the call option on maize? How much should Dr Evil pay the exchange that sold him the put option? The Black-Scholes equation is a mathematical model that gives a fair price for an option, given some assumptions. Although the Black-Scholes equation is no longer used for option pricing (it's been generalised to produce more sophisticated models that take many subtleties into account that it neglects) it still provides a good foundation for understanding option pricing.

Some notation before we get to the Black-Scholes equation:

t time, running from 0 (now) to T (expiry)

S the price of the underlying stock (or asset), $S \geq 0$.

V the price of the option, dependent on S and t

K the strike price, $K \geq 0$.

r the risk-free interest rate (e.g. interest on a US Treasury bond)

σ the standard deviation of the stock's return

The Black-Scholes equation is

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0.$$

We're not going to explain the derivation of the Black-Scholes equation here – it's not at all obvious why the equation above should give the price of an option. You'll just have to accept it for now, and in BUS4028F it should all be explained. The boundary conditions for the Black-Scholes equation depend on whether we are pricing a call or a put option. We discuss the European call option here.

The first boundary condition relates to the price at expiry. At the expiry time T , if the spot price S is less than the strike price K the call option has no value. It will be left to expire and the underlying stock can be purchased at market price. On the other hand, if the spot price S is greater than the strike price K at expiry then the option is "in the money" and it's worth $S - K$, the amount saved by purchasing at the

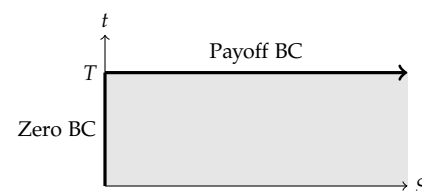


Figure 4.8: Black-Scholes domain for a call option

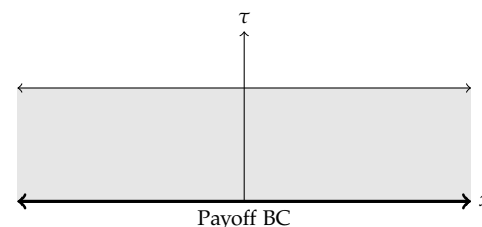


Figure 4.9: Transformed domain

strike price rather than the spot price. Together, this gives

$$V(S, T) = \max \{0, S - K\}.$$

The second boundary condition relates to price for a stock with zero value. A call option for this stock is worthless at any time, whatever the strike price since the underlying stock can be purchased for free on the market. This can be expressed as

$$V(0, t) = 0.$$

The Black-Scholes equation can be solved by making appropriate changes of variable that reduce it to the heat equation. Instead of presenting all of the transformations at once we perform three separate changes.

1. The first transformation is justified by the observation that if we ignore the term that involves a time-derivative the remaining terms look like a Cauchy-Euler equation.

$$\frac{\partial V}{\partial t} + \underbrace{\frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV}_{\text{Cauchy-Euler equation}} = 0.$$

This motivates the introduction of a new variable $x = \ln(\frac{S}{K})$.

We have taken the ratio of the spot price to strike price inside the logarithm (rather than just the spot price S) because this quantity is dimensionless. After substitution this yields

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 \frac{\partial^2 V}{\partial x^2} + \left(r - \frac{1}{2}\sigma^2\right) \frac{\partial V}{\partial x} - rV = 0.$$

2. The second transformation is of the time variable. The Black-Scholes equation has a boundary condition specified at some point in the future, $t = T$. We would prefer it if this was an initial condition. We therefore introduce a new variable τ defined by $\tau = \frac{1}{2}\sigma^2(T - t)$. We've also taken this opportunity to scale by the constant $\frac{1}{2}\sigma^2$ since this transforms the equation into

$$-\frac{1}{2}\sigma^2 \frac{\partial V}{\partial \tau} + \frac{\partial^2 V}{\partial x^2} + \left(r - \frac{1}{2}\sigma^2\right) \frac{\partial V}{\partial x} - rV = 0$$

which can be rearranged to give

$$\frac{\partial V}{\partial \tau} = \frac{\partial^2 V}{\partial x^2} + (k - 1) \frac{\partial V}{\partial x} - kV$$

where the dimensionless constant $k = \frac{2r}{\sigma^2}$.

3. The final transformation is of the function V . We assume that it has the functional form $V(x, \tau) = Ke^{\alpha x + \beta \tau} u(x, \tau)$. After taking partial derivatives (remembering to use the product rule) and cancelling out the common factor $Ke^{\alpha x + \beta \tau}$ from each term we find that

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2} + (2\alpha + k - 1) \frac{\partial u}{\partial x} + (\alpha^2 + \alpha(k - 1) - k - \beta)u.$$

The choice of $\alpha = -\frac{k-1}{2}$ and $\beta = -\frac{(k+1)^2}{4}$ reduces this to the heat equation

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}.$$

These transformations do not only change the differential equation, but also the boundary conditions of the differential equation. Again, we consider only the boundary conditions for the European call option. The payoff boundary condition $V(S, T) = \max\{0, S - K\}$ is transformed into an initial condition since $t = T$ corresponds to $\tau = 0$. More specifically

$$\begin{aligned} V(S, T) &= Ke^{-\frac{k-1}{2}x} u(x, 0) \\ \max\{0, S - K\} &= Ke^{-\frac{k-1}{2}x} u(x, 0) \\ \max\{0, K(e^x - 1)\} &= Ke^{-\frac{k-1}{2}x} u(x, 0) \\ \max\{0, e^x - 1\} &= e^{-\frac{k-1}{2}x} u(x, 0) \\ \Rightarrow u(x, 0) &= \max\left\{0, e^{\frac{k+1}{2}x} - e^{\frac{k-1}{2}x}\right\} \end{aligned}$$

Now that we have a heat equation on an infinite domain with an initial condition we can solve using the method outlined in the previous section. It's not difficult to find the solution but it is a long process. The basic ideas are outlined below along with some of the intermediate results so that you can check your calculations.

Some preliminary remarks. First, the variable k is slightly overloaded. We used it in this section to denote the constant $\frac{2r}{\sigma^2}$ and in the previous section as the constant in the heat equation. Since the Black-Scholes equation is transformed into the heat equation

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}$$

we won't have a problem. The variable k as it appears in equation 4.5 takes the value 1. Second, the initial condition

$$u(x, 0) = \max\left\{0, e^{\frac{k+1}{2}x} - e^{\frac{k-1}{2}x}\right\}$$

is equal to zero when $x \leq 0$ and it's positive for $x > 0$. Perhaps the easiest way to see this is to take out a common factor, i.e.

$$e^{\frac{k+1}{2}x} - e^{\frac{k-1}{2}x} = e^{\frac{k}{2}x} \left(e^{\frac{x}{2}} - e^{-\frac{x}{2}} \right).$$

The first term is always positive, and the second term (inside parenthesis) is an odd function that is negative when x is negative and positive when x is positive. With that in mind, we can proceed.

$$\begin{aligned} u(x, \tau) &= \int_{-\infty}^{\infty} u_0(s) \frac{1}{\sqrt{4\pi\tau}} e^{-(x-s)^2/4\tau} ds \\ &= \underbrace{\int_0^{\infty} e^{(k+1)s/2} \frac{1}{\sqrt{4\pi\tau}} e^{-(x-s)^2/4\tau} ds}_{I_1} - \underbrace{\int_0^{\infty} e^{(k-1)s/2} \frac{1}{\sqrt{4\pi\tau}} e^{-(x-s)^2/4\tau} ds}_{I_2} \end{aligned}$$

These two integrals are similar. We will discuss I_1 . I_2 can be solved in the same way. The idea is to multiply the two exponentials and then complete the square. This gives

$$I_1 = \frac{1}{\sqrt{4\pi\tau}} \int_0^{\infty} e^{-(s-(x+\tau(k+1)))^2/4t + (2xt(k+1) + \tau^2(k+1)^2)/4t} ds.$$

Now the change of variable $s = 2\sqrt{\tau}z + (x + \tau(k+1))$ transforms this integral to

$$I_1 = \frac{1}{\sqrt{\pi}} e^{(2xt(k+1) + \tau^2(k+1)^2)/4t} \int_{-(x+\tau(k+1))/2\sqrt{\tau}}^{\infty} e^{-z^2} dz.$$

This can be rewritten in terms of the complementary error function which is defined by

$$\text{Erfc}(w) = \frac{2}{\sqrt{\pi}} \int_w^{\infty} e^{-z^2} dz.$$

This gives the following expression for I_1

$$I_1 = \frac{1}{2} e^{(2x\tau(k+1) + \tau^2(k+1)^2)/4\tau} \text{Erfc} \left(-\frac{x + \tau(k+1)}{2\sqrt{\tau}} \right).$$

The integral I_2 has a similar expression

$$I_2 = \frac{1}{2} e^{(2x\tau(k-1) + \tau^2(k-1)^2)/4\tau} \text{Erfc} \left(-\frac{x + \tau(k-1)}{2\sqrt{\tau}} \right).$$

Now we take the solution $u(x, \tau) = I_1 - I_2$ and undo the transforms that converted the Black-Scholes equation into the heat equation.

First we write $V(x, \tau) = Ke^{\alpha x + \beta \tau} u(x, \tau)$ and after some useful cancellation of the exponentials we have

$$V(x, \tau) = \frac{K}{2} e^x \text{Erfc} \left(-\frac{x + \tau(k+1)}{2\sqrt{\tau}} \right) - \frac{K}{2} e^{-kt} \text{Erfc} \left(-\frac{x + \tau(k-1)}{2\sqrt{\tau}} \right).$$

The last step is to convert x, τ and k back to S, K, T, t, r and σ . Once we've done this we have

$$V(S, t) = \frac{S}{2} \text{Erfc} \left(-\frac{\ln(S/K) + (r + \sigma^2/2)(T-t)}{\sigma\sqrt{2(T-t)}} \right) - \frac{K}{2} e^{-r(T-t)} \text{Erfc} \left(-\frac{\ln(S/K) + (r - \sigma^2/2)(T-t)}{\sigma\sqrt{2(T-t)}} \right).$$

This result is often given in terms of the cumulative distribution function of the normal distribution

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-z^2/2} dz$$

and it's not difficult to convert between Erfc and Φ . If one does the solution becomes

$$V(S, t) = S\Phi\left(\frac{\ln(S/K) + (r + \sigma^2/2)(T - t)}{\sigma\sqrt{(T - t)}}\right) - Ke^{-r(T-t)}\Phi\left(\frac{\ln(S/K) + (r - \sigma^2/2)(T - t)}{\sigma\sqrt{(T - t)}}\right).$$

A representative solution is plotted in Figure 4.10 below.

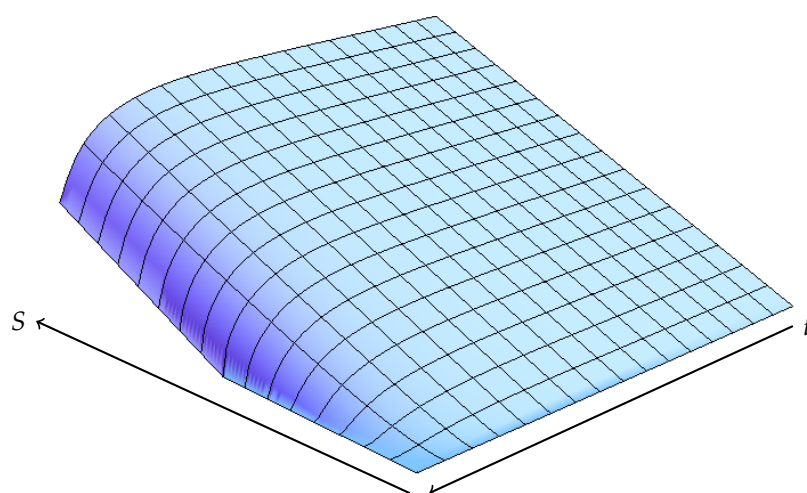


Figure 4.10: The solution to the Black-Scholes equation for a call option. Here $K = 2$, $T = 10$, $\sigma = 2$ and $r = \frac{1}{2}$. The S axis runs from 0 to 4 and the t axis runs from 0 to 10. Note the zero boundary value along $S = 0$ and the payoff boundary condition at $t = T = 10$.

Many of you will go on to have successful careers in finance, if not as actuaries then as financial analysts or hedge fund managers or some other highly-paid executive position where you'll be managing millions, perhaps billions, of dollars. Thinking is hard work, but you owe it to the people whose money you're managing to do it anyway.

"I believe the Black-Scholes formula, even though it is the standard for establishing the dollar liability for options, produces strange results when the long-term variety are being valued... The Black-Scholes formula has approached the status of holy writ in finance ... If the formula is applied to extended time periods, however, it can produce absurd results. In fairness, Black and Scholes almost certainly understood this point well. But their devoted followers may be ignoring whatever caveats the two men attached when they first unveiled the formula." – Warren Buffet